Unsupervised Domain Adaptation of Object Detectors: A Survey

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***Abstract*—Recent advances in deep learning have led to the development of accurate and efficient models for various computer vision applications such as classification, segmentation, and detection. However, learning highly accurate models relies on the availability of large-scale annotated datasets. Due to this, model performance drops drastically when evaluated on label-scarce datasets having visually distinct images, termed as domain adaptation problem. There are a plethora of works to adapt classification and segmentation models to label-scarce target dataset through unsupervised domain adaptation. Considering that detection is a fundamental task in computer vision, many recent works have focused on developing novel domain adaptive detection techniques. Here, we describe in detail the domain adaptation problem for detection and present an extensive survey of the various methods. Furthermore, we highlight strategies proposed and the associated shortcomings. Subsequently, we identify multiple aspects of the problem that are most promising for future research. We believe that this survey shall be valuable to the pattern recognition experts working in the fields of computer vision, biometrics, medical imaging, and autonomous navigation by introducing them to the problem, and familiarizing them with the current status of the progress while providing promising directions for future research.**

***Index Terms*—Deep learning, domain adaptation, object detection, transfer learning, unsupervised learning.**

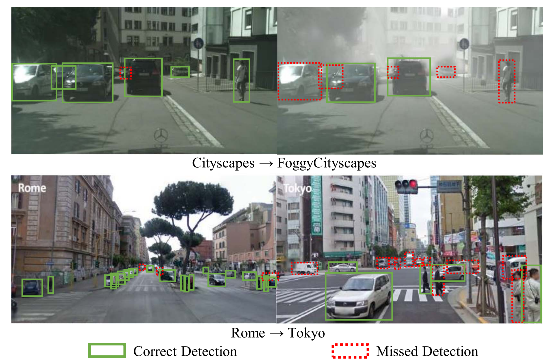

Fig. 1. Left: Source trained model on source domain, Right: Source trained model on the target domain. Top row: A detection model trained on the Cityscapes dataset, when evaluated on the FoggyCityscapes dataset, it fails to detect cars and pedestrians due to the domain shift caused by fog. Bottom row: A model trained in Rome, when evaluated on another city such as Tokyo, performs poorly due to differences in scene appearances, weather, objects, etc. These examples show that the detection models generalize poorly under the domain shift.

I. Introduction

THE success of deep learning has been greatly beneficial for various fields such as natural language processing [1], [2], [3], robotics [4], [5], [6], computer vision [7], [8], [9], etc. This is especially evident in the case of computer vision, where majority of the progress can be largely attributed to the advancements in deep convolutional neural networks (DCNN) [7]. Owing to their learning capacity, DCNN models have achieved state-of-the-art performance in many vision tasks such as object classification ([9], [10], [11]), semantic segmentation ([12], [13], [14]), and object detection ([15], [16], [17]). This has led to DCNN's increased popularity in several real world applications as compared to the classical computer vision techniques. Specifically, deep learning based object detection has become an integral part of many real-world applications ranging from video security/surveillance, augmented reality, autonomous navigation, human computer interface, self-checkout convenience stores. Major advancements like Faster-RCNN [15], You Only Look Once (YOLO) [16] and Single Shot Multi-box Detector (SSD) [17] have resulted in significant improvements of detection performance and speed.

It is important to note that most DCNN models need to be trained in a supervised fashion, which has been made possible due to the availability of large datasets having thousands of images annotated with ground-truth labels [18], [19], [20]. However, one of the major drawbacks is the poor generalization capability of DCNN models to visually distinct images compared to the training images. For instance, a detection model trained with a dataset collected in Rome may not necessarily perform well on images from Tokyo due to the changes in the appearance of scenes/objects and/or weather between them, as illustrated in bottom row of Fig. 1. A similar example is shown for cases such as sunny to foggy weather in Fig. 1 top row. Fig. 2 shows quantitatively the performance drop of different deep learning based object detectors that are trained on one particular dataset, when evaluated on different datasets. This problem where models, trained on one particular dataset (also known as source dataset), do not generalize well to a dataset that has a different distribution (also known as target dataset) is commonly referred to as *domain shift* or *distribution shift* in the literature [21], [22], [23], [24].

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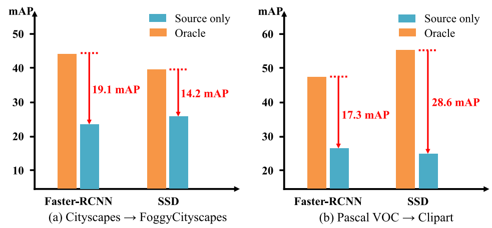

Fig. 2. Illustration of detector performance; In (a), the model is trained on Cityscapes and evaluated on FoggyCityscapes and in (b), the model is trained on Pascal VOC and evaluated on Clipart. We can observe a significant drop in the performance of the detector when there is a distribution shift in the training and test data.

A straightforward approach to solving this distributional shift problem is annotating the target dataset images with ground-truth detection labels. However, this might prove to be infeasible considering that the labor cost of the annotation process is prohibitively expensive for all visually distinct conditions. To circumvent this issue, many methods rely on the principles of unsupervised domain adaptation [21], [22] which involves training the DCNN model with both label-rich source dataset and label-scarce target dataset having visually distinct appearance. Techniques [24], [25], [26] for domain adaptive training with source and unlabeled target datasets have demonstrated improved generalization capabilities, resulting in improved performance on the visually distinct target domain. The unsupervised domain adaptation has been extensively studied for the task of classification [27], [28], [24], [29], [30], [31], [32], [33], [34], [35], [36], [37], [38], [39], [40], [41], [42], [43], and semantic segmentation [25], [44], [45], [46], [47], [48], [49], [50], [51], [50], [52], [53], [54], [55], [47], [56], [57].

However, unlike classification (image-level prediction) and semantic segmentation (pixel-level prediction), the object detection task involves bounding-box localization and the bounding-box-level category prediction task. This poses unique challenges while addressing unsupervised domain adaptation for object detection models. This has sparked an interest in addressing unsupervised domain adaptation for object detection with many novel approaches proposed very recently [26], [58], [59], [60], [61], [62], [63], [64], [65], [66], [67], [68], [69], [70], [71], [72], [73], [74], [75], [76], [77], [78], [79], [80], [81]. A timeline of some of the key papers proposed in the recent past is shown in Fig. 3.

While there exist multiple survey papers that extensively review domain adaptation techniques both classification [22], [82], [83] and semantic segmentation [84], [85], to the best of our knowledge, there is no comprehensive survey of unsupervised domain adaptation of object detectors. Although Li et al. [86] attempt to review some of the domain adaptive object detection literature, their discussions are limited and lack a comprehensive comparison of existing methods. This motivates us to present a comprehensive literature analysis of all the domain adaptive object detection methods proposed in the past few years, along with detailed discussions and comprehensive comparison. The major contributions of this work are summarized as follows:

1) As opposed to the existing survey [86], we provide a more detailed and thorough discussion on all the existing works (to the best of our knowledge) on domain adaptive object detection. We also define a taxonomy of the various works in literature. Furthermore, we discuss the preliminaries of the relevant topics like object detection and domain adaptation in an attempt to make the paper self-sufficient and useful to the readers who are not familiar with these concepts.
2) We present a comprehensive comparison of the existing methods on all publicly available datasets used in the literature for unsupervised domain adaptive object detection with thorough discussions on the methods and detailed comparison of the performances along with their respective experimental settings.
3) Finally, we identify potential research directions that might prove beneficial for the researchers working in the area in order to further advance the state-of-the-art.

II. Preliminaries

In this section, we provide an introduction to two of the important aspects related to domain adaptive object detection, i.e., object detection and unsupervised domain adaptation. We formally set up the problem and notations that are used throughout the paper.

A. Object Detection

Over the years, deep convolutional neural network based object detectors have demonstrated exceptional improvements in performance on a variety of datasets and have become an integral part of various computer vision applications. There are a variety of surveys [87], [88], [89] on the topic covering wide range of techniques proposed over the past decade for object detection. The most popular frameworks for object detection are Faster-RCNN [15], You Only Look Once (YOLO) [16] and Single Shot Multi-box Detector (SSD) [17]. The majority of domain adaptive object detection works are based on the Faster-RCNN and a few others use SSD. Other recent frameworks include, Fully Convolutional One Stage (FCOS) Object Detection [90] and DEtection TRansformer (DETR) [91]. However, these frameworks have been only scarcely used for the domain adaptive object detectors. In what follows, we briefly describe the some of the detection frameworks used in the domain adaptive detection literature, i.e., Faster-RCNN, SSD and DETR.

1) Faster-RCNN: The Faster-RCNN framework, proposed by Ren et al. [15], follows a two-stage object detection approach and it consists of three major components: *1)* a backbone CNN, *2)* a Region Proposal Network (RPN), and *3)* a Region-of-Interest (RoI) based classifier (RCN). Fig. 4(a) shows an overview of the Faster-RCNN architecture. Consider a dataset, $\mathcal{D} = \{X^i, Y^i\}_{i=1}^N$, having N images, with each image X^i with ground-truth annotation Y^i. Here, the ground-truth annotation Y^i denotes both bounding boxes and respective object categories in the corresponding image X^i. As shown in Fig. 4(a), an input image (X^i) is forwarded through the backbone network resulting in a set of feature maps. These feature maps are then fed to

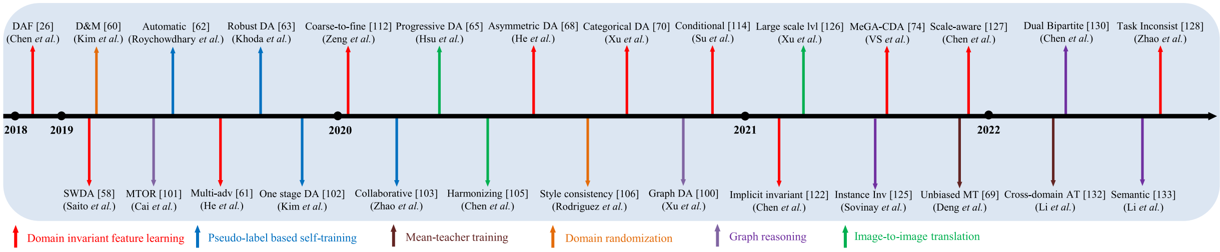

Fig. 3. A timeline of the key papers related to the domain adaptation of object detectors published over the recent years. These methods are broadly categorized into six classes: Adversarial feature learning, Pseudo-label based self-training, Graph-reasoning, Image-to-image translation, Domain randomization and Mean-teacher training.

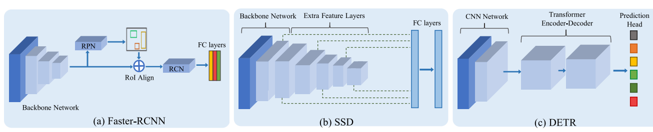

Fig. 4. Illustration of popular detection frameworks: (a) Faster-RCNN [15], (b) Single Shot Multi-Box Detector (SSD) [17].

the RPN network which generates a set of candidate object proposals. The RPN network relies on pre-defined anchor boxes of multiple sizes and aspect ratios in order to effectively learn to generate the candidate proposals. Subsequently, each proposal is then transformed into fixed-size features using RoI-pooling. Finally, the pooled features are then forwarded through the RCN, which predicts the category label for each candidate proposal in addition to refining its bounding box. For training the RPN candidate, a category-agnostic binary label (of being an object or not) is assigned to each anchor. The jth anchor is assigned a label, denoted as $y_b^j \in \{0, 1\}$, as positive (or 1) if it has the highest Intersection over Union (IoU) overlap with one of the ground-truth boxes or if it has an IoU overlap higher than 0.7 with any of the ground-truth boxes in the corresponding image. Similarly, a negative label (or 0) is assigned to the anchor if IoU ratio is lower than 0.3 for all ground-truth boxes. The RPN is then tasked to perform a binary classification to identify whether the candidate bounding box proposal corresponds to one of the objects in the image and learn the offset between the ground-truth bounding box, denoted as $\mathbf{b}^j$, and respective anchor box to get final bounding box prediction, denoted as $\tilde{\mathbf{b}}^j$. The offset learning is supervised with the help of a regression loss applied on the bounding box parameters. Both these losses are combined together to obtain the final loss for region proposal network as shown below:

$$\begin{aligned} \mathcal{L}_{rpn} &= \frac{1}{N_b} \sum_j \mathcal{L}_{rpn}^{bce}(y_b^j,\, p_b^j) \\ &\quad + \lambda_{rpn} \frac{1}{N_{bbox}} \sum_j p_b^j \mathcal{L}_{rpn}^{reg}(\mathbf{b}^j,\, \tilde{\mathbf{b}}^j), \end{aligned} \tag{1}$$

where j is the index of an anchor box in the mini-batch and p_b^j is the probability assigned to the respective anchor box being an object. The loss, $\mathcal{L}_{rpn}^{reg}$, computes the smooth-L1 distance between the given ground truth bounding box and the predicted bounding box $\tilde{\mathbf{b}}^j$. Next, the RCN network is trained to perform classification of RoI-pooled features using cross entropy loss with $K+1$ class classification, denoted as $\mathcal{L}_{rcn}^{ce}$. Here, K denotes the number of categories in the dataset and an additional class to represent the background category. Additionally, the RCN is also tasked to predict the bounding box offset through regression loss similar to the RPN network. The overall loss function used to train the entire Faster-RCNN network is trained is defined as:

$$\mathcal{L}_{det}^{frcnn} = \mathcal{L}_{rpn} + \mathcal{L}_{rcn}. \tag{2}$$

More details regarding the anchor boxes, regression losses, training, and architecture can be found in the [15].

2) Single Shot Multi-Box Detector (SSD): Liu et al. [17] proposed a single shot object detection framework which consists of forwarding the image through a single stage as opposed to two stages in the Faster-RCNN detector. Fig. 4(b) illustrates the SSD detection architecture. By following this approach, SSD eliminates the need for an object proposal stage, making it simpler and computationally efficient as compared to the Faster-RCNN approach. The SSD framework employs VGG16 as the backbone network which is used for extracting feature map of size $H \times W$ from an input image X. For each feature map location, SSD discretizes the output space of the bounding boxes into a set of default bounding boxes. A convolutional layer is added that for each feature map location predicts a score for a category or offsets relative to the default box coordinates. The set of default boxes contain bounding boxes of multiple pre-defined aspect ratios and scales to match any object shape in the image better. Furthermore, SSD combines predictions from feature maps at multiple scales to better handle the object scales with respect to the image. Once the model predictions are available, they are matched with the ground-truth box and category to perform an end-to-end training with regression ($\mathcal{L}_{reg}$) and classification ($\mathcal{L}_{cls}$) loss. The final detection loss is a combination of both regression and classification losses and is defined as follows:

$$\mathcal{L}_{det}^{ssd} = \mathcal{L}_{reg} + \mathcal{L}_{cls}. \tag{3}$$

In the case where there are no predicted bounding boxes that can be matched with one of the ground-truth bounding boxes, the regression loss is set to zero. More details regarding the default boxes, box matching algorithm bounding box regression losses, training procedure, and architecture details can be found in [17].

3) DEtection TRansformer (DETR): Carion et al. [91] proposed an object detection framework that views it as a set prediction problem and designed an object detector architecture termed as DEtection TRansformer (DETR). Carion et al. [91] argues that modeling capability of pairwise interaction in a sequence by transformer-based [1] architectures is specifically useful for set prediction. DETR consists of a DCNN-based backbone and a transformer-based encoder-decoder architecture similar to that of [1], which is trained in an end-to-end manner with a set loss function that performs bipartite matching between predictions and ground-truth. The overall architecture of DETR is illustrated in Fig. 4. For a single image pass, DETR provides a fixed size N_o (set larger to maximum possible objects in an image) object predictions. Here, N_o is set larger to maximum possible objects in an image. Let us consider, the ground-truth annotation Y denoting both bounding boxes and respective object categories in the corresponding image X. Let us consider predictions from DETR, denoted as $\hat{Y}$, as a set of N_o predictions. To perform bipartite matching between prediction and ground-truth, Y is padded with $\emptyset$, indicating no object. After computing matching cost between elements of ground-truth and predictions, the optimal prediction index is optimally computed based on Hungarian algorithm [91]. If we consider that a ground-truth element $(y, \mathbf{b}) \in Y$ has matched with a prediction $(p, \hat{\mathbf{b}}) \in \hat{Y}$, The Hungarian loss for the matched pair is then calculated as follows:

$$\mathcal{L}_{Hungarian}(y, \hat{y}) = \mathcal{L}^{ce}(p, y) + \mathbb{1}_{y \neq \emptyset} \mathcal{L}_{bbox}(\hat{\mathbf{b}}, \mathbf{b}) \quad (4)$$

Here, p is prediction probability for ground-truth category y, $\hat{\mathbf{b}}$ and $\mathbf{b}$ are predicted bounding box and ground-truth bounding box, respectively, $\mathbb{1}_{y \neq \emptyset}$ is an indicator function which is 0 when the matched ground-truth element in set Y is empty, i.e., $\emptyset$ and 1 otherwise. The bounding box loss $\mathcal{L}_{bbox}$ is a linear combination of generalized IoU loss [92] and $L1$ loss, i.e., $\lambda_{iou}\mathcal{L}_{iou} + \lambda_{L1}||\hat{\mathbf{b}} - \mathbf{b}||_1$. Where, λ_{iou} and λ_{L1} are hyperparameters and both these losses are normalized by number of objects inside a batch.

4) Popular Extensions of These Detection Frameworks: There are many other approaches that built on top of these frameworks to improve object detection performance in a supervised training setup. Some of the important such works are Mask-RCNN [93], Feature Pyramid Networks [94], and RetinaNet [95]. He et al. [93] proposed an extension of Faster-RCNN which added a small overhead of computing object mask as well on top of existing bounding box prediction head to improve the overall performance of the Faster-RCNN based object detector. The object mask computation is decoupled from the object class, i.e., a binary mask is predicted for each category. This additional branch in Mask-RCNN design helps it perform better compared to Faster-RCNN baseline benefiting from its multi-task training. Another extension of Faster-RCNN is proposed by Lin et al. [94], which proposed to incorporates Feature Pyramid Network (FPN) on top of the Faster-RCNN framework. Specifically, the proposed modification adds RPN overhead for feature map at each scale of the DCNN backbone, instead of using only the final feature map as proposed in vanilla Faster-RCNN [15]. The goal of this modification is to naturally leverage the feature hierarchy of backbone DCNN architecture to aid the object detection. With little computation overhead, the FPN enables detecting object at multiple scales resulting in improved detection performance. Lin et al. [95] proposed RetinaNet, which is one of the most popular detection backbone used in the literature in recent years. RetinaNet is a one-stage detector designed by borrowing aspects of other detection frameworks, e.g., using anchors similar to RPN of Faster-RCNN framework and feature pyramids similar to SSD or FPN. However, the major contribution of RetinaNet is their proposed balanced loss function, termed as focal loss [95], for training with highly imbalanced background/foreground classes. If we denote p as prediction probability for foreground class the focal loss can be defined as:

$$\mathrm{FL}(p) = -\alpha\, (1 - p)^{\gamma} \log(p), \quad (5)$$

where γ is a focal loss parameter that controls the weight on hard-to-classify examples for a highly skewed foreground:background ration case as in dense object detection (1:100). More specifically, the value of $\gamma > 1$ will assign low loss values for the easy samples and high loss for hard samples, thereby focusing on the hard samples while training. While, γ parameter controls the balance of easy/hard to classify examples, the α parameter balances the importance of positive/negative examples. Both the parameter balances the loss and helps the detector model train better on hard examples and improve detection and performs better compared to many other approaches.

B. Domain Adaptation

In the domain adaptation problem, we consider two domains, namely source and target, denoted as $\mathcal{S}$ and $\mathcal{T}$, respectively. The source and target domains are assumed to have different data distributions, i.e., $\mathrm{P}_{\mathcal{S}} \neq \mathrm{P}_{\mathcal{T}}$. Most domain adaptation formulations consider that the source dataset is label-rich, while the target dataset is label-scarce in nature [22]. Multiple variations of this formulation that are commonly studied in the literature include, *semi-supervised* [96], [97], [98], *weakly-supervised* [99], and *unsupervised* domain adaptation [21], [26], [58], [73], [74], [100], [101]. In the context of object detection, the *semi-supervised domain adaptation* formulation assumes that source domain is fully labeled with bounding box annotations and corresponding category labels and only a subset of the target domain samples are fully annotated with bounding box and respective category labels. *Weakly-supervised domain adaptation* formulation assumes that source domain is fully annotated and all target domain samples have binary annotations indicating the presence/absence of any category and no bounding box annotations. Lastly, the *unsupervised domain adaptation* formulation assumes that source domain is fully annotated while no annotations are available for target domain. Among these formulations, the unsupervised formulation is more practical and challenging. Further, the solutions obtained for this formulation can be easily adopted to address the semi-supervised and weakly-supervised domain adaptation tasks as well. For these reasons, we mainly focus on reviewing works that address unsupervised domain adaptation for object detection. In what follows, we formally define the unsupervised domain adaptation formulation and provide a brief overview of the same.

1) Unsupervised Domain Adaptation: Let us denote the source dataset as, $\mathcal{S} = \{X_s^i, Y_s^i\}_{i=1}^{N_s}$, and it consists of N_s number of images. Here, X_s^i denotes ith image and Y_s^i denotes the corresponding bounding box annotations with category label. Similarly, let us denote the target dataset as, $\mathcal{T} = \{X_t^i\}_{i=1}^{N_t}$ having N_t number of target domain images with no ground-truth annotations. Ben-David et al. [21] proposed a framework to perform domain adaptation for the given setup, i.e., labeled source dataset and unlabeled target dataset, with theoretical upper bounds on the target performance. Ben-David et al. [21] designed a $\mathcal{H}\Delta\mathcal{H}$-distance to measure the divergence between two sets of samples that have different data distributions, as is the case for the domain adaptation problem. Let us consider an arbitrary source domain image $X_s \in \mathcal{S}$ and an arbitrary target domain image $X_t \in \mathcal{T}$. Furthermore, let us consider a domain discriminator denoted as, $D : X \rightarrow \{0, 1\}$, that takes in any image $X \in \{\mathcal{S} \cup \mathcal{T}\}$ and predicts the domain of the input image. classifies source domain image $X_s \in \mathcal{S}$ as label 0, and target domain image $X_t \in \mathcal{T}$ as label 1. Considering $\mathcal{H}$ to be a set of possible domain discriminators, the $\mathcal{H}\Delta\mathcal{H}$-distance can be defined as follows:

$$d_{\mathcal{H}\Delta\mathcal{H}}(\mathcal{S}, \mathcal{T}) = 2 \sup_{(D,D')\in\mathcal{H}^2} \Big| \mathop{\mathbf{E}}_{X\sim\mathcal{S}}[D(X) \neq D'(X)] - \mathop{\mathbf{E}}_{X\sim\mathcal{T}}[D(X) \neq D'(X))] \Big|, \tag{6}$$

where $\mathbf{E}_{X\sim\mathcal{S}}$ and $\mathbf{E}_{X\sim\mathcal{T}}$ denotes the expected domain classification errors over the source and target domain dataset, respectively. More precisely, the (6) measures the divergence by the disagreement of the hypothesis sampled from $\mathcal{H}$. The ideal joint hypothesis is defined as:

$$D^* = \operatorname*{argmin}_{D\in\mathcal{H}} \mathrm{Err}_{\mathcal{S}}(D^*) + \mathrm{Err}_{\mathcal{T}}(D^*). \tag{7}$$

Here, the terms $\mathrm{Err}_{\mathcal{S}}$ and $\mathrm{Err}_{\mathcal{T}}$ denote the expected prediction errors on the source and target domain data samples, respectively. This distance is often used in the domain adaptation literature to measure the adaptability between any give source and target domain datasets. Ben-David et al. [21] present a theorem that further defines the upper bound on the target error as:

$$\forall D \in \mathcal{H},\ \mathrm{Err}_{\mathcal{T}}(D) \leq \mathrm{Err}_{\mathcal{S}}(D) + \frac{1}{2} d_{\mathcal{H}\Delta\mathcal{H}}(\mathcal{S}, \mathcal{T}) + Const. \tag{8}$$

We can note from the (8), the target error is upper bounded by three terms, namely expected prediction error on the source domain, domain divergence denoted in (6), and few constant terms. More details regarding both (6) and (8) are provided in [21]. A majority of the domain adaptation works in the literature rely on this formulation and focus on minimizing the upper bound on the target error by reducing the domain divergence between the source and target domain. In what follows, we discuss the different strategies used in the literature to address this specifically for the task of object detection.

III. Methods

As discussed earlier, a variety of methods [26], [58], [59], [60], [61], [62], [63], [64], [65], [66], [67], [68], [69], [70], [71], [72], [73], [74], [75], [76], [77], [78], [79], [80], [81] have been proposed for the task of domain adaptive object detection. Based on a meticulous review of these approaches, we categorize them into the following classes.

1) Domain Invariant Feature Learning: This class of adaptation approach performs an adversarial training of object detector model with the help of a domain discriminator. The training follows a gradient reversal layer based feature learning proposed by Ganin et al. [24]. This results in detector model producing domain invariant features which are useful to perform detection in target domain. Many methods in the literature utilize this strategy to adapt detectors to target domain [26], [58], [72], [73], [74].

2) Pseudo-Label Based Self-Training: Many works in the literature [62], [63], [99], [102], [103] utilize highly confident predictions by source-trained detector model to train it on the target. Since confident predictions on target domain have higher chance of being correct, such training strategy progressively makes detector model better on the target.

3) Image-to-Image Translation: These class of strategies utilize an unpaired image-to-image translation based model to map target image to a source-like image or vice versa. This reduces distribution shift in the visual domain and makes it easier for detector to perform well on the source-like target images. Many work in the literature utilize such approach for improving detector performance on the target domain [65], [104], [105], [106].

4) Domain Randomization: Another interesting way to improve the detector performance on target domain is to devoid all source-style bias from the model. Domain randomization strategy creates multiple stylized version of source domain data to train the detector model such that the model is not biased towards any one style and generalizes better on the target domain. Some works in the literature such as [60], [106] follow this strategy.

5) Mean-Teacher Training: Mean-teacher is an effective way to utilize unlabeled data to improve model generalization by progressively training a detector model in a student-teacher framework. This motivated a few works in the literature [69], [101] to explore mean-teacher training to adapt detector model by utilizing unlabeled target domain.

6) Graph Reasoning: Some works in the literature [100], [101], [125] exploit the inter-object and intra-object relationships that exist in detection dataset. These object relations are modeled through graphs which help the detector on target domain by training to enforce the same object relations. Fig. 5 illustrates the various categories of approaches. A comprehensive list of the existing approaches is presented in Table I. In what follows, we discuss the key papers related to the respective categories in the aforementioned list.

A. Domain Invariant Feature Learning

1) Adversarial Training Through Gradient Reversal: The adversarial feature learning is built on the theory proposed by

TABLE I
LIST OF DIFFERENT DOMAIN ADAPTIVE OBJECT DETECTION APPROACHES

Method	Detection framework	Type	Publication	Year
Faster-RCNN in the wild [26]	Faster-RCNN	Domain invariant feature learning	Chen *et al.*, CVPR	2018
Cross-domain weakly supervised adaptation [99]	SSD	Pseudo-label based self-training	Inoue *et al.*, CVPR	2018
Strong weak distribution alignment [58]	Faster-RCNN	Domain invariant feature learning	Saito *et al.*, CVPR	2019
Selective cross-domain alignment [59]	Faster-RCNN	Domain invariant feature learning	Zhu *et al.*, CVPR	2019
Diversify and match [60]	Faster-RCNN	Domain randomization, Domain invariant feature learning	Kim *et al.*, CVPR	2019
Automatic adaptation from unlabeled videos [62]	Faster-RCNN	Pseudo-label based self-training	Roychowdhury *et al.*, CVPR	2019
Mean teacher with object relations [101]	Faster-RCNN	Graph-reasoning, Mean-teacher training	Cai *et al.*, CVPR	2019
Multi-adversarial adaptation [61]	Faster-RCNN	Domain invariant feature learning	He *et al.*, ICCV	2019
Robust learning from noisy labels [63]	Faster-RCNN	Pseudo-label based self-training	Khodabandeh *et al.*, ICCV	2019
Self-training for one-stage detector [102]	SSD	Pseudo-label based self-training	Kim *et al.*, ICCV	2019
Multi-level adaptation [64]	Faster-RCNN	Domain invariant feature learning	Xie *et al.*, ICCV Workshop	2019
Pixel and feature adaptation [107]	Faster-RCNN	Image-to-image translation, Domain invariant feature learning	Shan *et al.*, Neurocomputing	2019
Adapting from synthesis to reality [108]	SSD	Domain invariant feature learning	Xu *et al.*, IEEE Access	2019
Improving localization [109]	Faster-RCNN	Domain invariant feature learning	Yu *et al.*, IEEE Access	2019
Cycle-consistent adaptation [104]	Faster-RCNN	Image-to-image translation	Zhang *et al.*, IEEE Access	2019
Cross-domain scene text [110]	Faster-RCNN	Domain invariant feature learning	Chen *et al.*, ICNIP	2019
Cross domain detection image translation [111]	Faster-RCNN	Image-to-image translation	Arruda *et al.*, IJCNN	2019
Graph-induced prototype alignment [100]	Faster-RCNN	Graph-reasoning	Xu *et al.*, CVPR	2020
Coarse-to-fine adaptation [112]	Faster-RCNN	Domain invariant feature learning	Zheng *et al.*, CVPR	2020
Harmonizing transferability and discriminability [105]	Faster-RCNN	Image-to-image translation, Domain invariant feature learning	Chen *et al.*, CVPR	2020
Cross-domain document object detection [113]	Faster-RCNN	Domain invariant feature learning	Li *et al.*, CVPR	2020
Categorical regularization [70]	Faster-RCNN	Domain invariant feature learning	Xu *et al.*, CVPR	2020
Prior-based detector adaptation [73]	Faster-RCNN	Domain invariant feature learning	Sindagi *et al.*, ECCV	2020
Every pixel matters [72]	FCOS [90]	Domain invariant feature learning	Hsu *et al.*, ECCV	2020
Collaborative training [103]	Faster-RCNN	Pseudo-label based self-training, Domain invariant feature learning	Zhao *et al.*, ECCV	2020
Conditional normalization network [114]	Faster-RCNN	Domain invariant feature learning	Su *et al.*, ECCV	2020
Spatial attention pyramid network [115]	Faster-RCNN	Domain invariant feature learning	Li *et al.*, ECCV	2020
Asymmetric tri-way training [68]	Faster-RCNN	Domain invariant feature learning	He *et al.*, ECCV	2020
Dual multi-label prediction [116]	Faster-RCNN	Domain invariant feature learning	Zhao *et al.*, ECCV	2020
One-shot cross-domain adaptation [117]	Faster-RCNN	Pseudo-label based self-training	D'Innocente *et al.*, ECCV	2020
Progressive adaptation [65]	Faster-RCNN	Image-to-image translation, Domain invariant feature learning	Hsu *et al.*, WACV	2020
Multi-scale robust discrimination [71]	Faster-RCNN	Domain invariant feature learning	Pan *et al.*, WACV	2020
Object detection via style consistency [106]	SSD	Domain randomization, Pseudo-label based self-training	Rodriguez *et al.*, BMVC	2020
Image-instance full alignment network [67]	Faster-RCNN	Domain invariant feature learning	Zhuang *et al.*, AAAI	2020
Free lunch for source-free adaptation [75]	Faster-RCNN	Pseudo-label based self-training	Li *et al.*, AAAI	2020
Forward-backward cyclic adaptation [118]	Faster-RCNN	Domain invariant feature learning	Yang *et al.*, ACCV	2020
Domain invariant region proposal [119]	Faster-RCNN	Domain invariant feature learning	Yang *et al.*, ICME	2020
Uncertainty-aware distributional alignment [120]	Faster-RCNN	Domain invariant feature learning	Nguyen *et al.*, ICM	2020
Region proposal oriented adaptation [121]	Faster-RCNN	Domain invariant feature learning	Alqasir *et al.*, ACIVS	2020
Cross-device OCT lesion detection [118]	Faster-RCNN	Domain invariant feature learning	Yang *et al.*, ISBI	2020
Memory-guided category-wise adaptation [74]	Faster-RCNN	Domain invariant feature learning	VS *et al.*, CVPR	2021
Implicit invariant one-stage network [122]	SSD	Domain invariant feature learning	Chen *et al.*, CVPR	2021
Unbiased mean-teacher [69]	Faster-RCNN	Mean-teacher training, Image-to-image translation, Domain invariant feature learning	Deng *et al.*, CVPR	2021
Domain-specific suppression [123]	Faster-RCNN	Domain invariant feature learning	Wang *et al.*, CVPR	2021
Augmented feature alignment [124]	Faster-RCNN	Image-to-image translation, Domain invariant feature learning	Wang *et al.*, TIP	2021
Instance-invariant progressive disentanglement [125]	Faster-RCNN	Domain invariant feature learning, Graph-reasoning	Wu *et al.*, TPAMI	2021
Large-scale instance-level image-to-image translation [126]	Faster-RCNN	Image-to-image translation, Domain invariant feature learning	Shen *et al.*, IJCV	2021
Scale-Aware Domain Adaptive Faster RCNN [127]	Faster-RCNN	Domain invariant feature learning	Chen *et al.*, IJCV	2021
Task-specific Inconsistency Alignment [128]	Faster-RCNN	Adversarial feature learning	Zhao *et al.*, CVPR	2022
Adaptive transformer-based detector [79]	DETR	Domain invariant feature learning	Zhang *et al.*, archived	—
Synergizing Adversarial Learning [129]	Faster-RCNN	Pseudo-label based self-training , Adversarial feature learning	Munir *et al.*, NIPS	2021
Dual Bipartite Graph Learning [130]	Faster R-CNN	Graph-reasoning	Chen *et al.*, ICCV	2021
Decoupled Cross-adaptation [131]	DETR	Adversarial feature learning	Jiang *et al.*, ICLR	2022
Cross-Domain Adaptive Teacher [132]	Faster R-CNN	Mean-teacher training, Adversarial feature learning	Li *et al.*, CVPR	2022
Semantic-complete Graph Matching [133]	Faster R-CNN	Graph-reasoning	Li *et al.*, CVPR	2022
Holistic and Hierarchical Aapatation [134]	Faster R-CNN	Adversarial feature learning	Xu *et al.*, CVPR	2022

Ben-David et al. [21] (see Section II-B1 for details). Specifically, the overall strategy involves minimizing the upper bound given in (8) by directly minimizing the $\mathcal{H}\Delta\mathcal{H}$-distance. As we can notice from $\mathcal{H}\Delta\mathcal{H}$-distance given in (6), this distance is inversely proportional to the error rate of the domain classifier D. The goal in a domain adaptation scenario is to reduce this distance, i.e., increase the domain classifier error. Ganin et al. [24] exploited this and proposed a novel gradient reversal approach to train any neural network model for domain adaptation. The overall goal is to achieve a domain invariant feature space of a backbone neural network with the help of a neural network-based domain classifier. Suppose we denote a domain classifier network as D and the backbone feature extractor network as F. In that case, the feature extractor network also tries to increase the domain classifier loss. The network F tries to minimize the task-specific loss (classification/segmentation/detection loss) and maximize

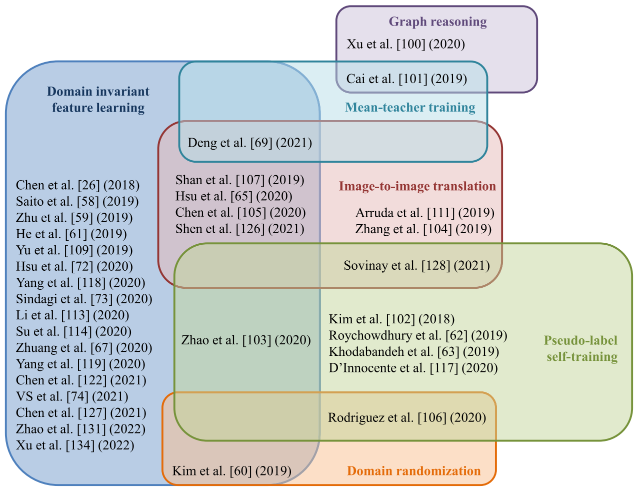

Fig. 5. The figure provides a Venn diagram of domain adaptive object detection methods. Each method falls into a class of adaptation approach (discussed in Section III), listed in the set representing respective techniques. Some of the work fall into more than one class of approaches, listed in the overlap region of set of respective class of adaptation techniques.

the domain classification loss in the overall training pipeline. The network D is trained to minimize domain classification loss. In addition to the task-specific loss, an additional loss involving domain classification is added. This loss is termed as adversarial loss [24] and it can be written as:

$$\max_{F} \min_{D \in \mathcal{H}} \left\{ \mathbf{E}_{\mathcal{S}}(D) + \mathbf{E}_{\mathcal{T}}(D) \right\}, \tag{9}$$

where $\mathcal{H}$ denotes the hypothesis space for the domain classifier and F is the feature extractor network. $\mathbf{E}_{\mathcal{S}}(D)$ and $\mathbf{E}_{\mathcal{T}}(D)$ denote the expected domain classification error over source and target domain, respectively. Eq 9 is implemented with the help of a gradient reversal layer which is applied before the input to the domain classifier. The gradient reversal layer during feed-forward acts as an identity function and the gradients are multiplied with -1 during backpropagation. In effect, this forces feature extractor F to maximize the domain classification loss while minimizing the task-specific loss resulting in the domain invariant feature space as proven by Ben-David et al. [21]. All the methods are utilizing this strategy to adapt a detector model using labeled source and unlabeled target domain fall under the *adversarial feature learning* category.

2) What are the Challenges of Domain Invariant Adversarial Training for Detection Compared to classification/segmentation?: The domain invariant feature learning through adversarial training was designed specifically for classification task [24]. Hence, it inherently assumes that each image contains one object semantic and the corresponding object-level feature would correspond to only one category. Furthermore, images of those category are often centered around the object minimizing influence of the background noise. However, this assumption holds only partially true for object detection task. For object detection only the classification module at the very end of the architecture where adversarial loss similar to [24] could be applicable. Application of adversarial training at that point can only be minimally useful. The domain-shift would

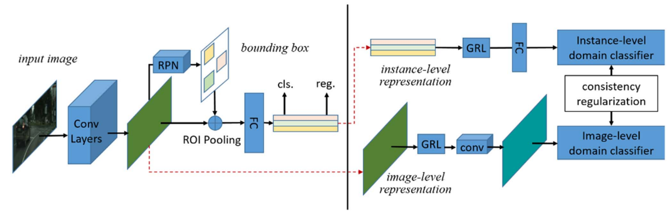

Fig. 6. Domain adaptive object detection in the wild [26] applies adversarial feature learning at image-level and instance-level with gradient reversal applied to detection backbone feature map and pooled features of the Faster-RCNN model. To regularize the adaptation, a consistency regularization is applied to make both image-level and instance-level domain classifier in sync with each other.

have been corrupted the features in multiple-levels of a detector pipeline, e.g., RPN, bounding-box regressor head etc for Faster-RCNN. Hence, an extension of domain adversarial training strategy is required that could overcome unique challenges posed by the object detection task. Specifically, adapting [24] inspired domain adaptive strategy would require to consider, semantically complex relationship of detector feature map that can contain information from multiple categories for a single image, avoiding influence of background noise, and focusing on feature alignment between same categories. Some of the early approaches like [26], [58] have proposed a multi-level alignment strategy for the object detector task by introducing adversarial feature alignment similar to [24] at multiple-levels of the object detector pipeline, which are discussed in detail in the Section III-A3. Following those early works, there are many other works (discussed in Section III-A4) which perform selective alignment of backbone feature maps using gating mechanisms to alleviate influence of background noise and encouraging alignment of semantically meaningful features.

3) Adaptive Detection Via Multi-Level Adversarial Training: Chen et al. [26] were among the first to formulate and address the problem of domain adaptive object detection. Given the problem setup discussed in Section II-B1 with a source $\mathcal{S}$ and target $\mathcal{T}$ domain, the method utilizes Faster-RCNN detection framework and proposes to utilize gradient reversal training at multiple stages of the detection framework. Specifically, given a backbone network F_b, RCN network F_{rcnn}, and RPN network F_{rpn} in the detection model, they apply adversarial training at both the *image-level* features which are extracted from backbone F_b and the *instance-level* features that are extracted from the RCN network, F_{rcnn}. Fig. 6 illustrates an overview of their approach. Let the features extracted from the backbone be denoted as $F_b(X) \in \mathbb{R}^{C \times H \times W}$, where C denotes the number of channels, H and W denote the height and width of the feature map, respectively. Furthermore, the discriminator networks for adversarial training at image-level and instance-level are denoted as D_{img} and D_{inst}. The image-level domain classification loss used to perform adversarial training at the image-level is then defined as:

$$\mathcal{L}_{img} = -\sum_{i,h,w} [y_d^i \log D_{img}(F_b(X^i)^{(h,w)}) + (1 - y_d^i) \log(1 - D_{img}(F_b(X^i)^{(h,w)}))], \tag{10}$$

where i indicates ith image in the batch, $h \in \{1, \ldots, H\}$, $w \in \{1, \ldots, W\}$, and $F_b(X^i)^{(h,w)}$ denotes the feature of size $1 \times C$ at location (h, w) in the feature map. The output of the discriminator network D_{img} is a probability score indicating whether the given image is from source domain or target domain. Here, $y_d^i \in \{0, 1\}$ denotes the domain label which is 0, when $X^i \in \mathcal{S}$ and 1, when $X^i \in \mathcal{T}$. Similarly, the instance-level domain classification loss used to perform adversarial training at the instance level is defined as:

$$\mathcal{L}_{inst} = -\sum_{i,j}[y_d^i \log D_{inst}(\mathbf{f}_{ij}^{pooled}) + (1 - y_d^i)\log(1 - D_{inst}(\mathbf{f}_{ij}^{pooled}))], \tag{11}$$

where $\mathbf{f}_{ij}^{pooled}$ indicates the RoI-pooled feature of size $1 \times d$ from the jth proposal of the image X^i. Since, both image and instance domain discriminators are trained independently and for any image X^i should result in same domain prediction, Chen et al. [26] introduce a regularization that enforces consistency across predictions of the domain discriminators. This consistency regularization is defined as:

$$\mathcal{L}_{consistency} = \sum_{i,j} \| \frac{1}{N_{act}^i} \sum_{h,w} D_{img}(F_b(X^i)^{(h,w)}) - D_{inst}(\mathbf{f}_{ij}^{pooled}) \|_2, \tag{12}$$

where N_{act}^i indicates number of activations in the feature map $F_b(X^i)$ of a given image X^i and $\| \cdot \|_2$ indicates $L2$-norm. Let us denote the overall detection model as $F = \{F_b, F_{rpn}, F_{rcnn}\}$. Combining all these loss functions, the final training objective is given as:

$$\max_{D_{inst},\, D_{img}} \min_{F} \mathcal{L}_{det}^{frcnn} - \lambda\, (\mathcal{L}_{img} + \mathcal{L}_{inst}), \tag{13}$$

$$\min_{F,\, D_{inst},\, D_{img}} \lambda\, \mathcal{L}_{consistency}, \tag{14}$$

where λ is a trade-off parameter used to balance adversarial and consistency loss. The detection and the discriminator networks are trained with final objective, (13) and (14). Saito et al. [58] argue that directly applying the gradient reversal at multiple levels in the backbone network is not necessarily optimal. Since shallower convolutional layers of the backbone capture local information, directly applying the adversarial loss would be beneficial in learning domain invariant local features between source and target domain data. This type of alignment is to as *strong alignment* in their work. Further, they reason that since the deeper layers in the backbone capture global information, performing a similar strong alignment would not be optimal. For example, when source and target domain data are sampled from different countries/cities, the number of objects in a scene, object co-occurrence, scene layout, etc can be very different. Consider the case when the source domain image contains only one object, whereas the target image contains multiple objects, performing strong alignment would likely increase the risk of misalignment. To tackle this issue, [58] modified the adversarial loss at the global-level (later convolutional layers of backbone network) by replacing the traditionally used binary cross-entropy loss with focal loss [95]. This strategy in [58] is referred to as *weak-alignment*. The overall approach then utilizes both the local *strong alignment* and global *weak-aliment* to reduce the domain gap between source and target domain data, resulting in increased detection performance for target images. Let the detection backbone network F_b be divided into two sub-networks that are cascaded together, namely global feature extractor F_g and local feature extractor F_l such that, $F_b(X) = F_g(F_l(X))$. The adversarial loss for strong alignment is defined as:

$$\begin{aligned}
\mathcal{L}_{local_s} &= \frac{1}{N_s HW} \sum_{i=1}^{N_s} \sum_{w=1}^{W} \sum_{h=1}^{H} \| D_l\left(F_l\left(X_s^i\right)\right)^{(h,w)} \|^2, \\
\mathcal{L}_{local_t} &= \frac{1}{N_t HW} \sum_{i=1}^{N_t} \sum_{w=1}^{W} \sum_{h=1}^{H} \| 1 - D_l\left(F_l\left(X_t^i\right)\right)^{(h,w)} \|^2, \\
\mathcal{L}_{local} &= \frac{1}{2}\left(\mathcal{L}_{local_s} + \mathcal{L}_{local_t}\right),
\end{aligned} \tag{15}$$

where D_l denotes the local domain discriminator, N_s and N_t are the number of source and target examples respectively, $\mathcal{L}_{local_s}$ and $\mathcal{L}_{local_t}$ denote the source and target domain classification loss respectively, and $F_l(X_s^i), F_l(X_s^i) \in \mathbb{R}^{C \times H \times W}$ are source and target local feature maps respectively. Denoting the global discriminator network as D_g, the weak alignment loss is defined as:

$$\begin{aligned}
\mathcal{L}_{global_s} &= -\frac{1}{N_s} \sum_{i=1}^{N_s} \mathrm{FL}(D_g\left(F_b\left(X_s^i\right)\right), \\
\mathcal{L}_{global_t} &= -\frac{1}{N_t} \sum_{i=1}^{N_t} \mathrm{FL}(1 - D_g\left(F_b\left(X_t^i\right)\right), \\
\mathcal{L}_{global} &= \frac{1}{2}\left(\mathcal{L}_{global_s} + \mathcal{L}_{global_t}\right),
\end{aligned} \tag{16}$$

where $\mathcal{L}_{global_s}$ and $\mathcal{L}_{global_t}$ are weak global alignment loss applied on source and target domain images, respectively. $F_b(\cdot)$ extracts the global feature, and $D_g(\cdot)$ denotes the probability of the global feature being from the source domain or target domain. FL denotes the focal loss [95].

4) Weighted Adversarial Feature Alignment: Most of the work in the domain adaptive detection literature that are based on adversarial feature learning focus on improving the gradient reversal training in order to obtain better feature alignment. Various methods such as [70], [72], [73], [74], [112], [116] build on the approaches discussed earlier and broadly follow the strategy of introducing a module that can control the gradient reversal information flow through loss weighting in addition to using a regularization technique to complement the proposed weighting module. For example, Zheng et al. [112] applies domain classifier at multiple levels of a detection backbone network to perform adversarial feature learning with gradient reversal layer. These adversarial losses are then multiplied by weights extracted from the backbone. In order to obtain the weights, the final feature map of the backbone is averaged over the channel dimension to obtain an attention map highlighting regions that potentially have an object. These weights are then used to modulate the adversarial loss. Similarly, He et al. [61] proposed a Multi-Adversarial Faster-RCNN (MAF) approach that extends the work by [26]. The first change in MAF is to apply image-level alignment at multiple layers of the backbone network. Additionally, the feature maps are reduced in the channel dimension to align the aggregated information rather than individual components. MAF also includes instance-level alignment but unlike [26], it weights the instance adversarial loss with the prediction probabilities produced by the RCN network. Xu et al. [70] a proposed categorical regularization strategy that can be combined with adversarial feature learning based approaches to further enhance the feature alignment between source and target domain. The categorical regularization strategy utilizes instance-level annotations from the labeled source domain data and adds a multi-label classification loss in addition to the detection losses. Such a strategy helps extract weak localization of objects in the feature maps through a multi-label classifier. The weak localization map is then used to gate image-level domain classification loss to block unnecessary information and highlight object regions.

Zhao et al. [116] combined the multi-label classification with the weak global alignment [58]. Specifically, a multi-label classifier is additionally trained along with the detection network with the help of source labeled data. The probability scores predicted by the multi-label classifier are used to condition the domain discriminator at the final layer of the backbone network. The conditioning mechanism takes in both the feature map extracted from the backbone and the multi-label probability vector indicating the probability of all K objects being in the image, using a multi-linear mapping function. Sindagi et al. [73] considered adverse weather conditions as a special case of domain adaptation. They argue that in the case of adverse weather, well-defined models are available that mathematically formulate how the camera captures such conditions. Using these models, they extract weather-specific information termed as "prior". These *priors* are then used to modify the conventional domain classification loss into a prior-prediction loss. In contrast to existing methods, Hsu et al. [72] do not utilize Faster-RCNN detection framework and instead they follow FCOS [90] one-stage detection framework that has centerness loss in addition to object classification and bounding box regression losses for supervised detection. The FCOS model provides a coarse prediction of center-focused heat-map indicating the location of the objects. They utilize this center-information to train the domain discriminators to perform center-aware feature alignment, rejecting any noisy information of the background. VS et al. [74] pointed out that existing adversarial learning-based approaches are prone to the category-wise *negative transfer* problem. When both source and target domain contains multiple categories, there are chances of misalignment across different categories of target and source features. Their key idea is to utilize $K+1$ category-specific domain discriminators for aligning each category separately. They introduce a memory module in the feature map extracted by the backbone network and produces attention maps that attend to the respective category features while blocking information from the rest of the categories. This ensures that the overall feature maps are aligned between source and target domain, while the additional category-specific discriminators prevent the negative transfer by performing the class-wise alignment.

Let F_b as the backbone network of the detection model, $X_s \sim \mathcal{S}$ and $X_t \sim \mathcal{T}$ be any arbitrary source and target domain image respectively, the typical weighted domain classification loss ($\mathcal{L}_{adv}^{weighted}$) used for most of the methods discussed in this section can be formulated as:

$$\begin{aligned}
&\mathcal{L}_{adv}^{weighted}(X_s, X_t) = \\
&\sum_{h=1}^{H}\sum_{w=1}^{W} y_d \cdot \sigma_{img_s}^{(h,w)} \cdot \left(1 - D\left(\sigma_{feat_s}^{(h,w)} \cdot F_b(X_s)^{(h,w)}\right)\right)^2 \\
&+ (1-y_d) \cdot \sigma_{img_t}^{(h,w)} \cdot \left(D\left(\sigma_{feat_t}^{(h,w)} \cdot F_b(X_t)^{(h,w)}\right)\right)^2, \quad (17)
\end{aligned}$$

where D is the image-level (in some cases instance-level) domain classifier, $y_d \in \{0,1\}$ is domain label which is 1 for source and 0 for target domain images, $\sigma_{img_t}, \sigma_{img_s} \in \mathbb{R}^{H\times W}$ are the image-level weights applied on the domain classification loss, $\sigma_{feat_t}, \sigma_{feat_s} \in \mathbb{R}^{H\times W}$ are the feature-level weights applied to mask-out irrelevant information and highlight important features that aid source and target domain alignment. Depending on the method, σ_{img} or σ_{feat} are set to identity, and either image-level or feature level weights are applied for the loss computations. Methods such as [61], [70], [73], [112], [116] utilize only σ_{img} weights to appropriately balance the adversarial loss. Whereas other methods such as [72], [74] apply only feature-level weights σ_{feat}. The contribution of each method comes from the way either σ_{img} and σ_{feat} is modeled. For example, Xu et al. [70] models σ_{img} with the help of multi-label classification probability vector and VS et al. [74] models σ_{feat} with category-specific memory modules.

5) Additional Methods: There are several subsequent works such as [56], [64], [67], [71], [77], [115], [119], [122], [125], [127] that utilize the adversarial feature learning strategies discussed in this section. Most of these approaches address cross-domain detection for the application of autonomous driving/surveillance. Some of the notable works that address the issue of domain shift in other applications include Yang et al. [118] which tackles the detector adaptation for medical tasks with the help of gradient reversal layer-based adversarial training for performing OCT lesion detection. In other tasks, Chen et al. [110] proposed adversarial feature training similar to [26] for adapting scene text detection models from synthetic data to outdoor settings, and Li et al. [113] utilized a multi-level feature alignment applied at multiple convolutional layers of the backbone network to train a detector for cross-domain detection on documents. Additionally, Li et al. [113] established a benchmark for the task of cross-domain detection in document space. Furthermore, there are many other interesting works available as pre-print [76], [81], [135], [136], [137], [138], [139]. Amongst different types of domain adaptive detection techniques, adversarial feature learning is the most popular approach with different variations available in the literature.

B. Pseudo-Label Based Self-Training

Self-training of object detectors on the target domain with pseudo label-based supervision is one of the simplest approaches to adapt the source-trained model to the target domain. Given a source pre-trained detection model, F^{src}, which is trained on the annotated source domain dataset $\mathcal{S} = \{X_s^i, Y_s^i\}_{i=1}^{N_s}$, it is used to generate pseudo-labels on the target domain data $\mathcal{T} = \{X_t^i\}_{i=1}^{N_t}$. The pseudo-labels, $\tilde{Y}_t^i = F^{src}(X_t)$, along with the images from the target dataset forms a new dataset, $\tilde{\mathcal{T}} = \{X_t^i, \tilde{Y}_t^i\}_{i=1}^{N_t}$. Here, $\tilde{Y}_t^i = F^{src}(X_t)$ denotes pseudo-labels obtained from a source-trained detector model F^{src} for ith target domain image X_t^i. Self-training typically involves using these pseudo-labels to re-train the network on the target data. In reality, the pseudo-labels are potentially noisy and often incorrect. Hence, directly training the model with these pseudo-annotations can potentially lead to a situation where the errors keep getting reinforced into the network. A filtering strategy is employed to overcome this issue, which involves removing annotations that are not confident. Most works follow two strategy for pseudo-label self-training. The first strategy is used to mine reliable pseudo-labels from video data through exploiting temporal consistencies or tracking algorithms and incorporating them for model training on target domain data. The second strategy involves developing complex and accurate pseudo-label filtering techniques to deal with the noisy pseudo-labels and avoid learning from them through loss re-weighting. The second strategy can be used together with the first, and especially when when it is not possible to access to video data.

Adaption Methods Prior to Deep Learning. This strategy of training with has been used even prior to the rise of deep neural networks in works such as Rosenberg et al. [140] and Tang et al. [141]. Rosenberg et al. [140] follow the later approach of filtering pseudo-labels through score and training the model. Specifically, Rosenberg et al. [140] extended a part based object detector model [142] learned in an unsupervised manner to a semi- and weakly-supervised setting which incrementally updates the training set with the help of newly labeled samples using model learned in the previous round. Whereas, Tang et al. [141] aims to adapt the object detector from image (source domain) to video (target domain) using the first strategy of mining pseudo-labels from video data through tracking algorithms. Specifically, Tang et al. [141] proposed a trajectory track selection approach, which is used to generate pseudo-labels to help the training on target domain. Furthermore, [141] decreases the amount of data from source domain and increases the target domain data to iteratively adapt the model.

Methods for Adapting DCNN-Based Detectors. One of the earliest approach for self-training of deep neural network based object detector for unsupervised domain adaptation proposed by Roychowdhry et al. [62] proposed to exploit the temporal information available in the video data similar to [141]. Specifically, they utilize a simpler strategy to exploit temporal consistency between adjacent frames to track detection by the source-trained model. This helps in mining more pseudo-labels which the detection model might have missed. Consequently, the pseudo-labels for the target dataset now contain predictions from the detection network and the tracker. It is trained on both labeled source and pseudo-labeled target dataset for adapting the detector to target data. The detector model is trained on both source and target domain data with both ground-truth and pseudo-labels. For any image X^i the detection loss is given as:

$$\mathcal{L}_{det}^i = \begin{cases} \mathcal{L}_{det}^{frcnn}(Y_s^i, p_s^i), & \text{if } X^i \in \mathcal{S} \\ \mathcal{L}_{det}^{frcnn}(\tilde{Y}_t^i, p_t^i), & \text{if } X^i \in \mathcal{T} \end{cases}, \tag{18}$$

where $\mathcal{L}_{det}^{frcnn}$ is Faster-RCNN detection loss, p_s^i and p_t^i are probability predictions from detector model, respectively for source and target, Y_s denotes annotation corresponding to respective source domain image that contains both bounding box and category label, and $\tilde{Y}_t$ denotes pseudo-label corresponding to respective target domain image.

Although mining pseudo-labels by exploiting the temporal consistency of video data is an effective strategy to obtain more annotations, it relies heavily on the availability of video data which is not always possible. Hence, it is important to develop single-image based pseudo-label training strategy. Khodabandeh et al. [63] proposed a method to specifically address the noise present in the pseudo-labels for the single-image target dataset. Their approach, termed as Robust Faster-RCNN, follows a three-phase training strategy. In the first phase, the detection network is trained with a supervised detection loss $\mathcal{L}_{det}^{frcnn}$ using a source domain dataset that has access to ground-truth annotations. In the second phase, the source-trained detector model is used to obtain pseudo-labels for the target domain dataset. Subsequently, the pseudo-labels are further refined using an additional classifier network that is pre-trained on a large-scale classification dataset. The refinement strategy utilizes both detector model prediction and classifier network predictions. With the help of refined labels, phase three involves training the detector model with a newly designed loss that accounts for potential noise in the pseudo-labels and helps the detector learn better.

Let us consider, $\mathbf{p}_c$ and $\mathbf{p}_c^{img}$ be the probability vector of a prediction from detector model and pre-trained image-classification network, respectively. Furthermore, let $\mathbf{s}_c$ and $\mathbf{s}_c^{img}$ denote the logits (score vectors) of a prediction from detector model and pre-trained image-classification network, respectively. Also, $\bar{\mathbf{b}}^{pseudo}$ and $\bar{\mathbf{b}}^{current}$ denote the bounding box pseudo-label collected in the first phase and current bounding box prediction by the detector model, respectively. The Robust Faster-RCNN [63] training utilizes these predictions to refine the pseudo-annotations, which are then used as supervision to train the detection model on the target data. The following equations describe the refinement process:

$$\begin{aligned} \mathbf{p}_c^{refined} &= \text{softmax}((\mathbf{s}_c + \alpha\, \mathbf{s}_{img})/(1+\alpha)), \\ \mathbf{b}^{refined} &= ((\tilde{\mathbf{b}}^{\,current} + \alpha\, \bar{\mathbf{b}}^{\,pseudo})/(1+\alpha)), \end{aligned} \tag{19}$$

where α is a hyper-parameter that controls the trade-off between the two terms. α starts with a large value and it is gradually decreased to smaller values as the third phase training progresses [63]. The theoretical formulation that lead to the refinement (19) can be found in [63]. Training with refined annotations help the detector model counter the noise in the pseudo-labels much better.

Previous methods to perform self-training were based on the Faster-RCNN framework and relied on subsequent assumptions to handle the noise in the obtained pseudo-labels. Kim et al. [102] proposed a strategy that focused on performing self-training of one-stage object detector models, specifically addressing it for the SSD-based model. The authors perform hard negative mining of pseudo-labels followed by a weak negative mining strategy. During the hard mining phase, a threshold δ is utilized to filter out the false-negative bounding box proposals based on their IoU with final bounding box prediction. Weak mining phase further improves the pseudo-labels by reducing the risk of false positives with the help of assigning instance-level scores calculated using Support Region-based Reliable Score (SRRS). The score is calculated by a set of RoIs predicted by the detection model that satisfies the hard mining threshold δ. If there are $N_{support}$ RoIs that satisfy the δ-IoU threshold corresponding to the any detector bounding box prediction $\tilde{\mathbf{b}}$ having confidence score of $\tilde{p}_b$, the SRRS score corresponding to the respective bounding box prediction can be calculated as:

$$\mathrm{SRRS}(\tilde{\mathbf{b}}) = \frac{1}{N_{support}} \sum_{i=1}^{N_{support}} \mathrm{IoU}(\mathbf{b}_i,\ \tilde{\mathbf{b}}) \cdot p_b. \tag{20}$$

Based on the SRRS scores, the pseudo-labels are further filtered with another threshold υ to reduce the false positives. Together with pseudo-labels mined through hard and weak mining phase, the detector model is trained in an end-to-end fashion. The self-training is performed by minimizing $\mathcal{L}_{det}^{ssd}$ as discussed in Section II-A2, where the loss is computed using the mined pseudo-labels.

Apart from these methods, there are many other interesting works available as pre-print [143], [144], [145] which utilize pseudo-label based self-training strategy for adapting detection model to target domain.

C. Image-to-Image Translation

As discussed earlier, the primary issue in unsupervised visual domain adaptation is that the target domain images are visually very distinct from the source domain. This causes a large gap in the feature space of the source-trained detector network, resulting in poor performance on the target domain. The method falling under adversarial feature learning (discussed in Section III-A) and pseudo-label based self-training (discussed in Section III-B) attempt to learn a feature representation that is more suited to perform detection on the target domain images. In contrast to feature alignment approaches, image-to-image translation-based methods to mitigate the domain gap at the input level. One of the most popular strategies used by image-to-image translation-based adaptation strategy is to use an unpaired image-to-image translation algorithm like Cycle-GAN [146], [147], UNpaired Image-to-image Translation [148], Multi-modal UNIT [149]. Methods that utilize the image-to-image translation-based approach often utilize additional strategies like self-training or adversarial feature learning to further boost the target detection performance. Image-to-image translation helps in bridging the gap at the input level while feature level adversarial alignment ensures a shared feature space for the target and source domain. Majority of state-of-the-art mapping methods do leave some image artifacts on the translated/generated images. For classification, images are much more low resolution compared to object detection and contain only one category. This helps learn image-to-image translation module with minimal artifacts in the images and benefits adaptation of for classification model using this strategy. However, for object detection learned image mapping module is likely to have more artifacts and lose part of the semantic information. Hence, most of the methods rely on a uni-directional image-mapping module that translates source images to target domain, as labeled information makes it possible to avoid any artifacts introduced by the image-mapping module during training. Due to complicated nature of image-to-image translation module for object detection, it is always used in conjunction with other type of methods such as domain adversarial training, mean-teacher framework, or extending any existing non-image-to-image translation methods by including an image-mapping module. For example, Zhang et al. [104] specifically utilize Cycle-GAN to learn a mapping function between source and target domain images, in a method termed as Cycle-consistent Adaptive Faster-RCNN (CA-FRCNN). Their approach extends the DA-Faster approach proposed in [26] (discussed in Section III-A3) by adding an image-to-image translation at the input level while adding other losses like gradient reversal at an instance and image-level intact. As shown in their experiments [104], the addition of an image-translation module further enhances the performance.

Hsu et al. [65] proposed a progressive adaptation strategy where they follow a similar strategy of using image-level translation. A Cycle-GAN based image-to-image translation module is utilized to translate target images into source-like images. Then with the help of the gradient reversal layer, the residual domain gap is further reduced in the feature space. Furthermore, with progressive adaptation, they showed that applying gradient reversal at only image-level is sufficient for adaptation rather than applying both image-level and instance-level losses. Arruda et al. [111] utilized the image translation module to specifically address the domain gap between daylight (source domain) and night-time (target domain) data. Once the image translation module is ready, it is used to translate all daylight data to create a fake night-time dataset. Since ground-truth annotations are available for the daylight images, they can be used to supervise the detector model with fake night-time images as input. Such training will help learn a detector model that is better suited for the night-time scenario as compared to a model trained with fully supervised daylight images.

Chen et al. [105] combined previously proposed strategies along with image-to-image translation in their approach termed as Harmonizing Transferability Calibration Network (HTCN). First, a Cycle-GAN based model learns a mapping between the source and the target domain. HCTN utilizes this image-translation module to create source-like target images and target-like source images, also termed as "interpolated images". The "interpolated images" helps HTCN fill in the distribution gap between domains and consequently reduce the source-bias of the decision boundaries. Additionally, HTCN employs local alignment loss similar to the one used in [58] for strong local alignment (discussed in Section III-A). Unlike other approaches based on image-to-image translation idea that directly utilize feature map outputs of a network, HCTN uses output of the local discrimination network to weigh the feature map of "interpolated images" forwarded to the subsequent layers to help avoid negative transfer while promoting positive transfer by appropriately weighting the feature maps. Furthermore, HCTN employs two additional discriminators to perform gradient reversal-based alignment of both masked features and global features at the output of the detection backbone network. By combining all of these losses, HCTN captures the transferable regions in the source, target and "interpolated image" domains to fully utilize the useful information for adapting images translated by Cycle-GAN.

Shen et al. [126] utilize MUNIT [149] based Image-to-image translation to extend the method proposed in [59] (discussed in Section III-A3). Specifically, MUNIT-based translation module is learned by learning to map both whole images (image-level translation) and object regions (instance-level translation) between source and target domains. Furthermore, they also learn a style code bank to obtain an object, background and global style vectors containing rich spatial information to aid the translation network. In addition to the MUNIT-based image translation module, discriminator networks are applied at multiple layers to perform gradient reversal-based adversarial training. Further, another set of domain classification networks are trained without any gradient reversal layer. The features learned by these domain classifiers are concatenated with RoI-pooled features to improve the classification. They also noted that detaching these domain classification networks to restrict the gradient flow from the main detection pipeline further improves the detection performance. Other interesting works [66], [78], [150], [151] utilizing image-to-image translation strategy can be found as pre-print.

D. Domain Randomization

In the case of image-to-image translation based methods, the primary focus is to learn a mapping between source and target domain and reducing the domain gap at the input-level. In case a perfect mapping function is available that can translate target domain images to source and vice versa, the detector model can perfectly adapt to target domain without requiring any annotations. However, in practice, such mapping function are not necessarily accurate and hence even after image translation, there might still exist a domain gap between original images and translated images, e.g., source images and translated target images or vice versa. Due to this, most image-to-image translation-based approaches utilize an additional gradient reversal based feature alignment or pseudo-label based self-training. To overcome this inability of learning accurate mapping function between source and target domain, domain randomization applies several transformations on the input image to derive multiple new domains. The goal for detection model is to then consider these new domains during training and learn feature representations that are invariant to all the domains including source and target. This strategy enforces strong constraints on feature representations of detection model as the network has to find features within images that remain invariant across a wide variety of image transformations. However, to achieve this, additional strategies such as adversarial feature learning or self-training are required.

In summary, image-to-image translation based methods aim to learn intermediate stages where input-level domain gap is reduced to improve model performance. Whereas, domain randomization synthetically generates domains that consist of multiple distinct styles (including source and target domain style) to ensure that the detection models learn features that are useful for detection regardless of style of the input images. Let us consider a source domain dataset $\mathcal{S} = \{X_s^i, Y_s^i\}_{i=1}^{N_s}$ and target domain dataset $\mathcal{T} = \{X_t^i\}_{i=1}^{N_t}$. Domain randomization creates multiple new datasets derived from the given source domain dataset that consist of multiple distinct styles. Let us denote these source-derived stylized dataset as, $\mathcal{S}^{style_m} = \{X_{s_m}^i, Y_s^i\}_{i=1}^{N_s}$, where $m \in \{1, \ldots, M\}$ denotes mth style and M denotes total number of unique styles. The stylization process ensures that the contents of the image are not changed and hence the object category and location information is preserved through the stylization process. All the stylized datasets are derived from source domain and hence, the ith stylized image $X_{s_m}^i$ can share the ground-truth annotations of corresponding ith image from the original source dataset X_s^i, for any $m \in \{1, \ldots, M\}$ and $i \in \{1, \ldots, N_s\}$.

Kim et al. [60] utilized domain randomization to adapt a Faster-RCNN based detection model. The method has three components: domain diversification module, detection model, and multi-domain discriminator. The domain diversification module is based on generative adversarial networks [152] and is tasked to take in the source domain images and shift the domain to derive a diverse set of visually distinct domains. Further, they enforce certain constraints on the output of domain diversification networks such as reconstruction, color preservation and cycle consistency. This ensures that the synthetically generated domains do not destroy the contents of the image that might negatively impact the model performance. Subsequently, the detection model is trained on these synthetically generated domains data along with the source and target domain data. Supervised detection loss, $\mathcal{L}_{det}^{frcnn}$, is applied on the source and synthetic domains to train the detector model. To ensure that the base network of detection model learns domain invariant feature representations, a multi-domain discriminator network with gradient reversal layer is employed at the end of the base network and is trained with a multi-class cross-entropy loss.

Rodriguez et al. [106] proposed a domain randomization approach for adapting SSD based one-stage object detectors. The key idea is to utilize the style transfer network proposed by [153] to create a source-derived dataset with multiple distinct predefined styles. Multiple stylized version of the source-derived dataset is created with annotations borrowed from the source domain dataset. Furthermore, the source-trained detector model is evaluated on the target domain images to extract pseudo-labels, which are then used for self-training. In order to extract robust pseudo-labels for effective self-training, a positive and negative threshold is utilized to extract high-quality positive and negative examples. The base network of the detector model is encouraged to learn domain invariant features through feature consistency loss that minimizes the $L2$-norm between feature representations extracted from source data and stylized source data. Apart from these methods, there is an interesting work available as pre-print [154], which also utilizes domain randomization approach for adaptation of object detectors.

E. Mean Teacher Training

Knowledge distillation has been demonstrated to be effective for exploiting unlabeled data in transfer learning [155], [156], [157], semi-supervised learning [158], [159], [160], and domain adaptation [161], [162], [163], [164]. Most of the domain adaptation work that utilize student-teacher training strategy has considered only the task of image classification. Their success has inspired many works which employ student-teacher training framework to perform unsupervised domain adaptation of object detector models. A recent and notable work based on this strategy is that of the unbiased mean-teacher strategy proposed by Deng et al. [69], that specifically utilizes mean-teacher framework [157] training for adapting object detector to the target domain. They combine multiple strategies like image-to-image translation (discussed in Section III-C) and adversarial feature learning (discussed in Section III-A) with mean-teacher framework. First, the method trains a Cycle-GAN module to learn image mapping between source and target domain images, which is then used to create a source-like target and target-like source images. The student model pipeline is trained with the original source domain, target domain and target-like source images. Since both source and target-like source images are fully labeled, they can be used for training using the supervised detection loss as source detection loss and target-like detection loss. Training the student pipeline with source and target domain images helps mitigate bias in the student model. Target-like source image training encourages student models to be more favorable towards target domain images. The model predictions of source-like target images are matched with model predictions of original target domain images to perform knowledge distillation. Further, the teacher parameters are updated with Exponential Moving Average as:

$$\Theta^{i}_{\mathcal{F}tch} = \alpha\Theta^{i-1}_{\mathcal{F}tch} + (1-\alpha)\Theta^{i}_{\mathcal{F}stu}, \tag{21}$$

where $\Theta^{i}_{\mathcal{F}tch}$ and $\Theta^{i}_{\mathcal{F}stu}$ are network parameters for teacher and student model respectively, α denotes smoothing coefficient hyper-parameter that can be used to control the teacher updates, the super-script i and $i-1$ denote the indices for the current and previous training iterations respectively. To further decrease the domain gap in the feature space of the student model, gradient reversal-based adversarial training involving strong local and weak global feature alignment [58]. Together with distillation, parameter updates supervised detection loss and adversarial feature training; the entire training is performed in an end-to-end fashion. Other works that utilize the student-teacher framework include Liu et al. [160] which extended the mean-teacher framework to train an object detector in a semi-supervised fashion by exploiting large unlabeled data to improve the overall performance. Though Liu et al.focuses on semi-supervised detection with labeled and unlabeled data drawn from the similar distributions, it can be easily adopted to perform unsupervised domain adaptive training of object detectors. Cai et al. [101] proposed cross-domain detection by utilizing a mean-teacher framework that relies on learning object relations through graph structures. Another interesting work [165], available as a pre-print, utilizes mean-teacher framework is also available as pre-print.

F. Graph-Reasoning

Graph-reasoning is a widely popular technique in the computer vision community for visual recognition [166], [167], video understanding [168], [169] etc. It is also utilized extensively for unsupervised adaptation of classification networks [170], [171]. The ability of graphs to model inter-image and intra-image relationships of underlying object categories can be extremely useful in the case of adapting object detectors. Furthermore, modeling graph structures is mutually exclusive and hence can be combined with other existing adaptation strategies. Unlike UDA classification where graph networks are used to model hidden structures of the data [170], for object detection each image sample would contain multiple object and inter-relation structure between them could be explicitly modeled using graph relations, especially through source domain label information. Leveraging these intern-object relation is uniquely possible for the object detection task and it provides helpful information to aid adaptation by maintaining those object relations across domains. Motivated by these benefits, Cai et al. [101] proposed a domain adaptive object detection approach, termed as Mean-Teacher Object Relations (MTOR), that utilizes graph-reasoning based strategy combined with student-teacher training. MTOR consists of four major components: *1)* Regional-Level Consistency (RLC), *2)* Inter-Graph Consistency (InterGC), *3)* Intra-Graph Consistency (IntraGC), *4)* Mean-Teacher Training (MTT). Here, the first three components are devoted to graph structures that model relationship between the objects within an image. Each vertex in the graph structure corresponds to a region proposal and is assigned a probability vector denoting the probability of that region belonging to one of the categories. Before selecting the region proposal, they are filtered by eliminating all proposals having a maximum probability value falling below a threshold.

To compute the inter-graph consistency, InterGC, the affinity matrix of both student and teacher graph is calculated using cosine similarity. Once we have both student and teacher model affinity matrix for a given target domain image, the model is trained to maintain the inter-graph consistency between source and target domain. This inter-graph consistency enforces alignment of the graph structure between student and teacher pipeline for a target image. Within each image, there can be multiple categories that are semantically related. Intra-graph consistency reinforces the similarity between proposals of the same category for student graphs with supervision from the teacher. Specifically, the teacher model is used to predict whether two region proposals contain an object of the same category. In addition to these graph-based consistency losses, the student network is trained with supervised detection loss. The teacher network parameter updates follow mean-teacher formulation discussed in Section III-E.

Recently, Xu et al. [100] proposed a graph-based adaptation approach where they construct a relational graph between object proposals to compute category prototypes that help alignment between source and target domain. The training strategy involves graph construction and alignment. The proposed method is based on the Faster-RCNN detection framework and utilizes its two-stage detector training to compute relational graphs. The relational graphs are used to compute category-wise prototypes, which are then matched to enforce alignment between source and target domain at each stage of the detector model. The overall alignment strategy is termed as Graph-induced Prototype Alignment (GPA) in the paper [100]. Each node in the graph represent a region proposal predicted by the detection model and the adjacency matrix values are computed by calculating Intersection over Union (IoU) between pairs of region proposals. Once the relation graph is constructed, the adjacency matrix is used to aggregate the RoI-pooled features and their respective class probabilities for each region proposals. Furthermore, the updated features and probabilities are used to calculate category-wise prototypes. The alignment is performed by matching source and target prototypes by minimizing the distance between prototypes of the same category while maximizing the distance between different categories. Additionally, the same category prototypes are matched between source and target domain by minimizing respective category prototypes.

IV. Evaluation Protocols

In this section, we discuss the details of the evaluation protocol used in evaluating the performance of the various domain adaptive detection approaches discussed earlier. First, we provide details of various benchmark datasets that are used for the evaluation purpose, followed by a discussion of various settings such as adaptation to adverse weather conditions, adaption to real-world data from synthetic data, etc. Next, we define the metrics used for evaluation and comparison. Finally, we provide a detailed discussion of the results of various methods.

A. Datasets

A variety of datasets have been used for evaluating the domain adaptive object detection approaches. Most of these datasets have been extensively used for evaluating object detection approaches, and hence, can naturally be used for the purpose of evaluating domain adaptive methods by simply defining different sets of data as source and target domain. These datasets can be classified into the following categories: (1) General objects [19], [99], (2) Self-driving [172], [173], [174], [175], (3) Face detection [176], [177], (4) Weather degradation [174], [175], [176], [178], and (5) Synthetic datasets [179]. Table IV provides summary of various datasets train and test samples.

Cityscapes. Scene understanding of complex urban streets is an important problem statement for a wide range of applications. To address this issue, the Cityscapes dataset was released in 2016 by a group of researchers in Daimler and TU Dresden [173]. The Cityscapes dataset contains 2975 images for training and 500 images for testing.

FoggyCityscapes. Analyzing scenes of urban streets under adverse weather is a challenging problem. To consider such scenarios, the FoggyCityscapes was introduced [175], by applying a fog filter over the Cityscapes dataset.

Sim10 K. Considering that the collection of real-world data is time-consuming and tedious, advancements in computer graphics have been exploited to generate photo-realistic data which can be easily rendered and annotated. With this motivation, Sim10K [179] was released which is rendered by the gaming engine Grand Theft Auto.

KITTI. KITTI [172] dataset contains hours of videos of traffic scenarios recorded with various high-quality sensors like RGB, grayscale and depth sensors. The KITTI object detection dataset consists of 7,481 training images and 7,518 test images, comprising of 80,256 labeled objects.

Pascal VOC2007/2012. PASCAL VOC [19] is a real-world dataset containing general objects. The widely used 2007 and 2012 version contains training and validation split with a total of 15 k images.

Clipart, Watercolor, Comic. Clipart [99], Watercolor [99] and Comic [99] datasets contain abstract, artistic and comical images, respectively. Understanding these abstract-type images will allow for the direct study of how a model infer high-level semantic information. The Clipart contains a total of 1 K images with 20 categories same as Pascal VOC. The watercolor and comic dataset contain 6 categories and contain 1 K training and 1 K testing images.

BDD100 K. The BDD100 K dataset [174] is an autonomous driving dataset contains large diversity in terms of scenes, geographical regions. It is the largest (till-date) driving video dataset with 100 K videos containing scenes from New York, Berkeley, San Francisco and Bay Area. The dataset contains 70 k training images and 10 k validation images. For domain adaptative detection, a subset containing 36,728 training and 5,258 validation daytime images.

WIDER FACE. Shuo [177] released the largest benchmark dataset consisting of 32,000 images and 199 K labeled faces. The dataset contains high degree of variability in scale, expression, illumination, pose, makeup and occlusion.

UFDD. The UFDD dataset [176] consists of images collected under a variety of weather conditions. The dataset capture the following conditions: rain, snow, haze, lens impediments, blur, illumination and contains a total of 6424 images.

RTTS. REalistic Single Image DEhazing (RESIDE) [178] is a large-scale dataset consisting of both real and synthetic hazy images. RTTS is an object detection dataset and is a subset of the RESIDE. It contains 4,807 un-annotated and 4,322 annotated real-world hazy images covering mostly traffic and driving scenarios.

B. Adaptation Scenarios

In this section, we describe the various adaptation scenarios and protocols followed by the existing approaches.

1) Adverse Weather Conditions: Stable detection performance in different weather conditions is important for safety-critical applications like self-driving cars. Weather conditions introduce image artifacts which can negatively impact the detection performance. FoggyCityscapes and Cityscapes can be utilized as target and source domains to evaluate the effectiveness of adaptation methods in adverse weather. Moreover, under the real hazy condition, one can extend the domain adaptation setting for Cityscapes to the RTTS dataset, used as the source and target domain, respectively. Similarly, for face detection, such weather conditions prove challenging when the face detector is trained with clean weather data. This adaptation scenario can be explored with the source domain as WIDER-Face and target domain as UFDD-Haze.

2) Synthetic Data Adaptation: Synthetic data offers an inexpensive alternative to real data collection as it is easier to collect, and with appropriate engineering, annotations can be made readily available for synthetic data with very little labor cost. In spite of the advancements in computer graphics, photo-realistic synthetic data generated using state-of-the-art rendering engines suffer from subtle image artifacts, which can result in sub-optimal performance on real-world data. The adaptation from Sim10 K (source domain) to Cityscapes (target domain) is used in the literature to analyze this setting.

3) Cross-Camera Adaptation: Differences in the intrinsic and extrinsic camera properties like resolution, distortion, orientation, location result in images that capture the objects differently from each other in terms of quality, scale, and viewing angle. While the collected data can be real, these domain differences will potentially result in severe performance degradation. To evaluate the domain adaptation methods under cross-camera adaptation settings, most literature considers KITTI to Cityscapes and KITTI to Cityscapes.

4) Adaptation Between Dissimilar Domains: In the shift from real-world to artistic images, underlying features, the texture of real-world images completely differ in artistic images. Hence, a lot of methods consider adaptation of dissimilar domains with three adaptation settings: PASCAL-VOC to Clipart, PASCAL-VOC to Watercolor and PASCAL-VOC to Comic. All these domain shifts are from real-world images to synthetic as well as artistic images.

5) Adaptation to Large-Scale Dataset: The increasing availability of cheap and good quality cameras has made collecting large-scale datasets much easier. However, annotating them for the detection task is labor-intensive. Therefore, exploring the possibility of adopting from a smaller dataset to a larger dataset has significant real-world implications. To evaluate this adaptation setting, methods in most domain adaptation literature consider Cityscapes to BDD100 k. In the BDD100 K dataset, only the daylight subset is considered as the target domain and the Cityscapes dataset as the source domain.

C. Discussion of Results

In this section, we provide detailed comparison and discussion of the performance of various domain adaptive object detection methods. For comparison, we only consider adaptation scenarios used by two or more works in the literature. Most of the approaches conduct adaptation experiments that consist of various scenarios created by carefully selecting a source and target domain from the datasets discussed in the previous section. To indicate an adaptation scenario, we follow dataset1$\rightarrow$dataset2 format, where dataset1 is used as a labeled source domain and dataset2 is used as an unlabeled target domain.

The performance comparison reported in Table II uses the absolute mAP performance as reported in the respective papers. It is important to note that this comparison is not necessarily fair for the following reasons: Each method uses different version of the baseline detector. This is true even for the methods that use Faster-RCNN framework. These methods use different hyperparameters for training like the number of epochs, learning rate, etc. Hence, to provide a fair, comprehensive comparison, we consider relative mAP improvement, denoted as ΔmAP (reported in Table III), which is the difference between the performance of a particular approach and that of the corresponding source-only baseline as reported in the respective paper. Since all methods in the literature train both source-only baseline and final method with same hyperparameters, ΔmAP would obviate most of the influences on the performance caused by hyperparameters of the respective methods.

Comparing these two tables provide some interesting insights into the performance trends over the years. For example, consider the case of Cityscapes $\rightarrow$ BDD100 K, where, from Table II, it seems that the forward-backward cyclic adaptation [118] approach improves over the progressive adaptation [65] strategy. However, considering the performance in terms of ΔmAP, both these methods obtain similar performance improvements. This suggests that the Cityscapes $\rightarrow$ BDD100 K scenario needs to be explored further by future works. The case of adverse weather adaptation case (Cityscapes $\rightarrow$ FoggyCityscapes), and has observed $\sim$20 mAP improvement since the first work [26]. From the absolute mAP comparison in Table II, the transformer-based adaptation [79] and the memory-guided adaptation [74] provide the best and second-best performance. Whereas from the ΔmAP comparison, it can be observed that the unbiased mean-teacher [69] and harmonizing transferability [105] obtains the most improvement over their respective source-only baseline. A similar trend is observed in other adaptation scenarios as well. Note that for Sim10 K $\rightarrow$ Cityscapes, conditional normalization [114] has second-best performance for both mAP and ΔmAP.

1) Is One Category of Adaptation Methods Better Than the Other Category of Approaches?: After discussing all the class of methods (i.e., adversarial feature learning, pseudo-label based self-training etc) and comparing their performances, it is natural to wonder if one class of methods is better than the rest.

From Tables II and III, we can see that there is no clear winner among different class of methods. Each class of adaptation strategy has its benefit and drawbacks. For example, adversarial feature learning (discussed in Section III-A based training is known to be unstable in practice and requires careful parameter tuning and regularization to stabilize the feature adaptation process. However, under appropriate hyperparameters and with correct regularization, adversarial feature learning can provide

TABLE II
QUANTITATIVE COMPARISON (MAP) OF EXISTING DOMAIN ADAPTIVE OBJECT DETECTION METHODS

Method	**CS → F**	**S10K → CS**	**KI → CS**	**CS → KI**	**VOC → Clip**	**VOC → WC**	**VOC → Comic**	**CS → BDD**	**Framework**	**LR**	**Additional Comments**
Faster-RCNN in the wild [26]	27.6	38.9	38.5	64.1	-	-	-	-	FRCNN	0.001	SS: 500
Diversify and match [60]	34.6	-	-	-	41.8	52.0	34.5	-	FRCNN	0.001	SS: 600
Multi-adversarial adaptation [61]	34.0	41.1	41.0	72.1	-	-	-	-	FRCNN	0.001	-
Progressive adaptation [65]	36.9	-	43.9	-	-	-	-	24.3	FRCNN	0.001	-
Strong weak distribution alignment [58]	34.3	40.7	-	-	38.1	53.3	-	-	FRCNN	0.001	SS: 600
Mean teacher with object relations [101]	35.1	-	-	-	-	-	-	-	FRCNN	0.001	SS: 600
Large-scale instance-level [126]	40.3	-	-	-	-	-	-	-	FRCNN	0.001	SS: 600
Curriculum self-paced learning† [180]	-	47.6	43.8	-	37.8	-	-	-	FRCNN	0.001	SS: 600
Image-instance full alignment network [67]	36.2	47.1	-	-	-	-	-	-	FRCNN	0.001	SS: 600
Forward-Backward cyclic adaptation [118]	36.7	42.7	-	-	38.5	53.6	-	-	FRCNN	0.001	SS: 600, Epochs: 10
Categorical regularization [70]	37.4	-	-	-	38.3	-	-	26.9	FRCNN	0.001	SS: 600
Collaborative training [103]	35.9	44.5	43.6	-	-	-	-	-	FRCNN	0.001	SS: 600
Robust learning from noisy labels [63]	36.4	42.5	42.9	77.6	-	-	-	-	FRCNN	0.001	SS: 600
Unbiased mean-teacher [69]	41.4	43.1	-	-	42.7	56.9	-	-	FRCNN	0.001	SS: 600
Prior-based DA Object Detection [73]	39.3	-	-	-	-	-	-	-	FRCNN	0.001	SS: 600
Coarse-to-fine adaptation [112]	38.6	43.8	-	41.0	-	-	-	-	FRCNN	0.001	SS: 600
Dual Multi-Label Prediction [116]	38.8	-	-	-	-	56.0	33.5	-	FRCNN	0.001	SS: 600, Epochs: 10
Instance-invariant progressive disentanglement [125]	36.4	-	-	-	42.1	56.9	-	-	FRCNN	0.001	SS: 600
Graph-induced prototype alignment† [100]	39.5	47.6	47.9	-	-	-	-	-	FRCNN	0.001	SS: 600, Epochs: 20
Harmonizing transferability [105]	39.8	42.5	-	-	40.3	-	-	-	FRCNN	0.001	SS: 600
Asymmetric Tri-way Faster-RCNN [68]	38.7	42.8	42.1	73.5	42.1	54.9	-	-	FRCNN	0.001	SS: 600
Selective cross-domain alignment [59]	33.8	43.0	42.5	-	-	-	-	-	FRCNN	0.0001	SS: 512, Epochs: 25
Spatial attention pyramid network [115]	40.9	44.9	43.4	75.2	42.2	55.2	-	-	FRCNN	0.00001	SS: 600, Iter: 90K
Cycle-consistent adaptation [104]	33.2	41.5	41.7	-	-	-	-	-	FRCNN	0.002	SS: 600
Multi-level adaptation [64]	36.0	42.8	-	-	-	-	-	-	FRCNN	0.002	SS: 600, Epochs: 10
Multi-scale robust discrimination [71]	37.0	43.4	-	-	-	-	-	-	FRCNN	0.002	SS: 600, Iter: 90K
Domain invariant region proposal [119]	39.2	45.5	44.0	-	-	-	-	-	FRCNN	0.002	SS: 600, Epochs: 10
Memory-guided category-wise adaptation [74]	41.8	44.8	43.0	75.5	-	-	-	-	FRCNN	0.002	SS: 600, Epochs: 10
Conditional normalization network [114]	36.6	49.3	44.9	-	-		-	-	FRCNN	0.00625	SS: 512, Epochs:12
Self-training for one-stage detector [102]	-	-	-	-	35.7	49.9	26.8	-	SSD	0.001	SS: 300, Iter: 120K
Implicit invariant one-stage network [122]	-	-	-	-	37.8	51.5	30.1	-	SSD	0.001	SS: 300
Object detection via style consistency [106]	34.3	-	-	-	44.8	57.3	39.4	-	SSD	0.00001	SS: 300, Epochs: 20
Every pixel matters [72]	36.0	49.0	43.2	-	-	-	-	-	FCOS	0.005	SS: 800, Epochs: 10
Adaptive transformer-based detector [181]	43.5	55.3	-	-	-	-	-	-	DETR	0.0002	Epochs: 50

CS: Cityscapes, F: FoggyCityscapes, S10K: Sim 10K, KI: KITTI, VOC: Pascal VOC, Clip: Clipart, WC: Watercolor, BDD: BDD100K, SS: Shorter side, LR: Learning Rate, Iter: Iteration, FRCNN: Faster-RCNN. Red and blue color indicate best and second-best methods in respective adaptation scenario in terms of mAP. † denotes that the corresponding method uses ResNet-50 backbone in the detection model, rest of the methods utilize VGG16 backbone.

TABLE III
QUANTITATIVE COMPARISON (ΔMAP) OF EXISTING DOMAIN ADAPTIVE OBJECT DETECTION METHODS

Method	**CS → F**	**S10K → CS**	**KI → CS**	**CS → KI**	**VOC → Clip**	**VOC → WC**	**VOC → Comic**	**CS → BDD**	**Framework**	**LR**	**Additional Comments**
Faster-RCNN in the wild [26]	8.80	8.85	8.30	10.6	-	-	-	-	FRCNN	0.001	SS: 500
Diversify and match [60]	16.7	-	-	-	16.9	12.2	13.1	-	FRCNN	0.001	SS: 600
Multi-adversarial adaptation [61]	15.2	11.0	10.8	18.6	-	-	-	-	FRCNN	0.001	-
Progressive adaptation [65]	17.3	-	15.1	-	-	-	-	3.50	FRCNN	0.001	-
Strong weak distribution alignment [58]	14.0	6.10	-	-	10.3	8.7	-	-	FRCNN	0.001	SS: 600
Mean teacher with object relations [101]	8.20	-	-	-	-	-	-	-	FRCNN	0.001	SS: 600
Large-scale instance-level [126]	20.0	-	-	-	-	-	-	-	FRCNN	0.001	SS: 600
Curriculum self-paced learning† [180]	-	16.9	12.3	-	11.7	-	-	-	FRCNN	0.001	SS: 600
Image-instance full alignment network [67]	15.2	12.0	-	-	-	-	-	-	FRCNN	0.001	SS: 600
Forward-Backward cyclic adaptation [118]	17.9	11.5	-	-	10.7	9.00	-	-	FRCNN	0.001	SS: 600, Epochs: 10
Categorical regularization [70]	15.4	-	-	-	11.3	-	-	3.50	FRCNN	0.001	SS: 600
Collaborative training [103]	9.70	10.0	8.7	-	-	-	-	-	FRCNN	0.001	SS: 600
Robust learning from noisy labels [63]	4.5	11.5	11.8	21.4	-	-	-	-	FRCNN	0.001	SS: 600
Unbiased mean-teacher [69]	19.6	8.80	-	-	13.6	8.00	-	-	FRCNN	0.001	SS: 600
Prior-based DA Object Detection [73]	14.9	-	-	-	-	-	-	-	FRCNN	0.001	SS: 600
Coarse-to-fine adaptation [112]	17.8	8.80	-	7.60	-	-	-	-	FRCNN	0.001	SS: 600
Dual Multi-Label Prediction [116]	15.4	-	-	-	-	11.4	13.8	-	FRCNN	0.001	SS: 600, Epochs: 10
Instance-invariant progressive disentanglement [125]	13.7	-	-	-	14.3	12.3	-	-	FRCNN	0.001	SS: 600
Graph-induced prototype alignment† [100]	12.6	13.0	10.3	-	-	-	-	-	FRCNN	0.001	SS: 600, Epochs: 20
Harmonizing transferability [105]	19.5	7.90	-	-	12.5	-	-	-	FRCNN	0.001	SS: 600
Asymmetric Tri-way Faster-RCNN [68]	18.4	8.20	11.9	20.0	14.3	10.3	-	-	FRCNN	0.001	SS: 600
Selective cross-domain alignment [59]	7.60	9.10	5.10	-	-	-	-	-	FRCNN	0.0001	SS: 512, Epochs: 25
Spatial attention pyramid network [115]	17.6	10.3	13.2	21.7	14.4	10.6	-	-	FRCNN	0.00001	SS: 600, Iter: 90K
Cycle-consistent adaptation [104]	8.78	6.70	4.00	-	-	-	-	-	FRCNN	0.002	SS: 600
Multi-level adaptation [64]	13.2	8.50	-	-	-	-	-	-	FRCNN	0.002	SS: 600, Epochs: 10
Multi-scale robust discrimination [71]	17.1	9.90	-	-	-	-	-	-	FRCNN	0.002	SS: 600, Iter: 90K
Domain invariant region proposal [119]	16.6	11.3	9.30	-	-	-	-	-	FRCNN	0.002	SS: 600, Epochs: 10
Memory-guided category-wise adaptation [74]	17.4	10.5	12.8	22.0	-	-	-	-	FRCNN	0.002	SS: 600, Epochs: 10
Conditional normalization network [114]	10.5	15.0	7.80	-	-	-	-	-	FRCNN	0.00625	SS: 512, Epochs:12
Self-training for one-stage detector [102]	-	-	-	-	9.00	2.80	4.90	-	SSD	0.001	SS: 300, Iter: 120K
Implicit invariant one-stage network [122]	-	-	-	-	11.1	4.4	8.20	-	SSD	0.001	SS: 300
Object detection via style consistency [106]	8.70	-	-	-	14.5	7.70	14.5	-	SSD	0.00001	SS: 300, Epochs: 20
Every pixel matters [72]	17.2	18.9	13.0	-	-	-	-	-	FCOS	0.005	SS: 800, Epochs: 10
Adaptive transformer-based detector [181]	9.5	4.80	-	-	-	-	-	-	DETR	0.0002	Epochs: 50

CS: Cityscapes, F: FoggyCityscapes, S10K: Sim 10K, KI: KITTI, VOC: Pascal VOC, Clip: Clipart, WC: Watercolor, BDD: BDD100K, SS: Shorter side, LR: Learning Rate, Iter: Iteration, FRCNN: Faster-RCNN. Red and blue color indicate best and second-best methods in respective adaptation scenario in terms of ΔmAP. † denotes that the corresponding method uses ResNet-50 backbone in the detection model, rest of the methods utilize VGG16 backbone.

TABLE IV
SUMMARY OF VARIOUS DATASETS USED IN DOMAIN ADAPTIVE OBJECT DETECTION EXPERIMENTS

Dataset	Train		Test	
	Images	*Catagories*	*Images*	*Catagories*
Cityscapes	2975	8	500	8
VOC-2007	5011	20	5011	20
VOC-2012	11540	20	11540	20
SIM10K	10000	1	10000	1
KITTI	7481	1	7481	1
FoggyCityscapes	2975	8	500	8
Clipart	1000	20	1000	20
Watercolor	1000	6	1000	6
BDD100K	36728	10	5258	10
Comic	1000	6	1000	6
WIDER FACE	32000	1	32000	1
UFDD	442	1	442	1
RTTS	4807	5	4322	5

All the datasets except RTTS contain labeled training images and test images. The RTTS dataset contains unlabeled training images and labeled test images.

significant performance improvements. Most likely for this reason, the most popular strategy used in the literature is adversarial feature learning-based methods [26], [58], [70], [72], [74], [182]. Self-training with pseudo-labels (discussed in Section III-B has a stable training curve as supervised detection loss can be used to train the network with supervision from pseudo-labels obtained through the source-trained model. However, self-training methods need to be careful of the noise in the pseudo-labels as such it can potentially get re-enforced into the model resulting in sub-optimal performance.

Image-to-image translation based methods (discussed in Section III-C try to avoid such training related issues by addressing the domain gap in the input-space. Typically, a image-translation module is added that transforms source samples to target-like samples and vice versa. Since target-like samples will have supervision readily available, the model can be trained better to perform detection on original samples from the target domain. However, this strategy heavily relies on the efficacy of the image-translation module to learn a perfect mapping between source and target domain. For this reason, most image-to-image translation methods are used in addition to the other strategies [65], [104], [105]. Domain randomization (discussed in Section III-D) based approaches on the other hand, create multiple stylized version of the source domain in order to ensure that the detection model is devoid of style-bias from source domain and focus on other important discriminative features to detect the objects. However, to completely de-bias the detector from any specific domain, many random domains need to be included during training, which might not be feasible. To address this, domain randomization methods also additionally apply adversarial feature learning between source, stylized-source, and target domain. Mean-teacher training (discussed in III-E) is another popular strategy used to adapt detectors to a target domain through mean-teacher updates [157]. Mean-teacher [157] has been successfully used for transfer learning and was initially proposed to utilize unlabeled data of similar domains. These approaches utilize additional regularization or feature matching strategy to extend the mean-teacher for unsupervised domain adaptation [69], [101]. Similarly, graph-reasoning (discussed in III-F) based methods aim to understand object relations within any given sample of the source domain and try to leverage that information to make sure the target domain also follow similar relations. This helps the knowledge transfer between source and target domain by enforcing certain constraints on the target features. Most of these different categories of approaches are mutually exclusive and can be combined in different ways to leverage the benefits of each strategy while mitigating the drawbacks of other strategies.

V. PROMISING RESEARCH DIRECTIONS

As discussed in the earlier sections, various methods have been proposed to address the challenging problem of adapting deep object detectors to different scenarios. In this section, we explore outstanding issues along with potential solutions and future research directions. In the following subsections, we discuss two broad categories of issues and directions: (i) Comprehensive evaluation and (ii) Improving generalization with real-world constraints.

A. Comprehensive Evaluation

We identify some of the issues concerning the evaluation protocol. We believe that addressing these issues would result in a more robust and rigorous evaluation process.

- *Generalizing to other detection frameworks*: Most of the existing domain adaptation approaches for object detection are evaluated only on the Faster-RCNN detection framework [15]. However, considering the popularity of single shot detection approaches such as YOLO [16], SSD [17], and more recent ones such as FCOS [90] and DETR [91], it is important to evaluate the performance of the adaptation methods using other frameworks as well.
- *Evaluation on complex real-world datasets*: Majority of domain adaptive object detection methods have been evaluated on the datasets such as FoggyCityscapes, Sim10 K, *etc* following the protocol designed by the initial approaches. However, these datasets carefully curated and do not reflect the real-world distribution gap. For example, consider the case of building a self-driving system where it is important to construct source and target datasets such that the distribution gap arises from a variety of factors such as differences in weather conditions, geographical locations, types of vehicles, types of backgrounds, densities of vehicles, *etc*. BDD100K [174] is one such complex dataset that reflects many such real-world constraints, but it has not been widely adopted for performance evaluation. Similar to works like [62], [65], [70], we encourage that future works include evaluation on datasets like BDD100 K which include many challenging real world conditions for the detection task. Reporting performance on such real-world dataset is crucial to enable a more rigorous and robust evaluation process.
- *Considering other applications*: Existing methods in the literature mainly focus on applications such as autonomous

navigation [26], [60], [72], [74], crowd surveillance [73], and commonly found general objects [58], [68], [70]. Whereas multiple other real world applications are rarely explored in the context of domain adaptive detection, such as medical imaging [118], scene text [110], document objects [113]. For example, consider a lesion detection model trained on Optical Coherence Tomography images captured from a particular device.

B. Improving Generalization With Real-World Constraints

In this section, we focus on potential future research directions that consider some real-world settings typically ignored in the existing approaches. Most of the existing methods are based on assumptions that need not necessarily be true in the real-world. For example, it is often assumed that (i) the number of samples across classes in both source and target domain dataset is assumed to be balanced, (ii) all the classes existing in the source domain dataset are present in the target domain dataset and vice versa, (iii) source domain dataset is always available during target domain adaptation, (iv) all target domain samples are unlabeled, and (v) large number of unlabeled target domain samples are available at the training. We discuss various problem settings where such assumptions are relaxed, the resulting challenges and possible strategies to overcome them. Furthermore, we also discuss additional strategies such as multi-source domain adaptation, test-time adaptation, and continuous adaptation necessary for real-world practice settings.

- *Weak/semi-supervised domain adaptation*: Unsupervised domain adaptation is specifically useful because annotating the target domain samples is labor-intensive and costly for the object detection task. However, in some scenarios, it might be possible to obtain bounding-box annotations for a subset of target samples or provide weak image-level annotations indicating the presence/absence of the categories. In such cases, this additional knowledge can be leveraged to improve the performance on target domain further. For weakly supervised adaptation, each image is annotated at image-level to indicate which categories are present in the image and no bounding box annotations are provided [99]. A subset of the target dataset is labeled with bounding-box and respective category annotation for semi-supervised adaptation [160].
- *One/few-shot domain adaptation*: Conventional unsupervised domain adaptation setting assumes availability of a large number of unlabeled samples from the target domain. However, this might not hold true in many realistic conditions where data is an incoming stream of images [117] resulting in either one or few unlabeled samples available from the target domain during adaptation training. To address this sample scarcity in the target domain-specific adaptation strategy is needed to get the best out of the available dataset [117]. A straightforward strategy to address one/few-shot adaptation for object detectors would be to apply image stylizing to increase the pool of unlabeled target domain samples and apply a conventional feature-matching strategy by making detector domain invariant [183].
- *Imbalanced classes*: Object frequency in the real world often follows a power law, and this imbalance class problem is typically ignored by existing domain adaptive detection methods which assume aligned class spaces. Hence, it is important to avoid the dominance of any particular class while performing the domain alignment. An obvious solution to this problem is to re-weight the classes based on the frequency [47], [184].
- *Partial/open-set domain adaptation*: A related problem to the class imbalance issue is that of partial or open-set domain adaptation. A typical assumption in the domain adaptation literature is that the label space between the source and target dataset is same. It might often be the case that the target dataset has much fewer classes than the ones in the source dataset, aka, partial adaptation. Another case might be that the target domain contains unknown classes that are not observed in the source domain, aka, open-set adaptation. Aligning the source domain completely with the target domain might result in negative transfer in such cases. Hence, it is important to mitigate the problem of negative transfer by down-weighting the data samples from classes that are not common between the domain, thereby promoting feature alignment only in the shared label spaces (positive transfer) [185], [186], [187], [188], [189].
- *Source-free domain adaptation*: A common assumption in the existing literature is that the samples from the source domain are available during domain adaptation training. However, in real-world scenarios, gaining access to source data might not be practical due to privacy concerns, legal issues, and inefficient data transmission. An obvious approach to address this issue would be to perform pseudo-label based self-training. However, it is important to generate highly reliable pseudo-labels by filtering out incorrect labels [75], [190], [191], [192], [193], [194], [194].
- *Multi-source domain adaptation*: All the existing approaches for domain adaptive object detection assume that the labeled training data are sampled from a single domain. This neglects a more practical scenario where training data are collected from multiple sources. Aligning the target data with all these sub-distributions might not necessarily result in optimal performance. Hence, it would be essential to perform a selective adaptation where the only relevant source samples are considered [195], [196].
- *Continuous and test time adaptation*: Existing domain adaptive detection approaches consider the world to be separated into stationary domains. However, this may not be the case in real-world, where samples arise from a continuously evolving underlying process. This setting reflects a real-world scenario where we encounter unlabeled images from new target environments that are not observed during training. Hence, it is important to develop strategies to adapt at test time where we have access only to a single or a few instances of target domain data [197], [198], [199], [200]. Note that these strategies should consider additional constraints such as computationally inexpensive

adaptation, restricted access to source domain data, and the issue of catastrophic forgetting, which is a prominent issue that results from continuous learning [171], [201], [202], [203], [204].

VI. Conclusion

In this paper, we considered the unsupervised domain adaptation of deep object detectors and presented an extensive survey for this task. We have reviewed several approaches that were published in the last few years. We have provided a comprehensive taxonomy of the existing approaches, followed by a detailed analysis of the various methods with their merits and demerits. We have also provided a detailed discussion on existing datasets, evaluation protocols and comprehensive performance comparison. Finally, we have identified some of the outstanding issues and promising directions to drive future research.

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